

Enhanced Elimination of Poisons

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CHAPTER OUTLINE

OVERVIEW OF CORPOREAL TREATMENTS FOR ENHANCED ELIMINATION OF POISONS, 1920

PRINCIPLES AND FACTORS INFLUENCING POISON REMOVAL DURING EXTRACORPOREAL TREATMENTS, 1921

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KEY POINTS

- Treatment of poisonings includes aggressive supportive care and consideration of antidotes, gastrointestinal decontamination, and elimination enhancement techniques.
- Characteristics that favor elimination of a poison with extracorporeal treatments include a small molecular size, low volume of distribution, low protein binding, and low plasma clearance.
- The anion and osmol gaps are crucial in the diagnosis and management of toxic alcohol poisonings.
- Of all the extracorporeal treatments available, hemodialysis is the most available, least expensive, most efficient (in terms of blood clearance), and associated with the lowest incidence of complications.
- For almost all cases for which an extracorporeal treatment is indicated, intermittent hemodialysis is the preferred option.

Poisonings are a major cause of morbidity and mortality. According to the National Poison Data System compiled by the American Association of Poison Control Centers, more than 2 million human poisonings were reported in the United States in 2021.¹ The classes of poisons accounting for the majority of fatal cases were sedatives, hypnotics, antipsychotics, cardiovascular agents, opioids, stimulants, and other recreational drugs. Table 66.1 shows the 2021 U.S. statistics of the incidence and outcomes related to various poisons that may require the care of a nephrologist.

All acute exposures to potential poisons should be considered life-threatening until a complete risk assessment has been performed.² The general approach to the poisoned patient necessitates prompt resuscitation and stabilization, clinical and laboratory evaluation, antidote administration, and gastrointestinal decontamination if appropriate. Antidotes can act at a receptor (e.g., naloxone for opioid poisoning), inhibit metabolism (e.g., fomepizole for toxic alcohol poisoning), or bind to inactivate the poison (e.g., deferoxamine for iron poisoning). Some antidotes

can be removed by extracorporeal treatments (ECTRs), such as ethanol or fomepizole, requiring an increased dose during ECTR, or may need to be repeated, as in the case of idarucizumab in patients with severe chronic kidney disease (CKD). Table 66.2 illustrates some of the most common antidotes used in the toxicology setting.

A minority of poisonings can benefit from treatments that actively enhance elimination. Activated charcoal (50 g in adults) is the preferred method of decontamination³; it is usually administered within 2 hours of ingestion and may be repeated for modified release and enteric-coated drugs. Contraindications include an altered level of consciousness with an unprotected airway, seizures, protracted vomiting, and impaired intestinal function (obstruction or ileus). Complications are rare and include aspiration, appendicitis, and bowel obstruction. Whole bowel irrigation removes poisons from the intestines (e.g., administration of a polyethylene glycol [macrogol] electrolyte solution [1–2 L/hour]) until the rectal effluent is clear. Whole bowel irrigation may be considered in massive ingestions of modified release drugs (e.g., salicylate), poisons not

Table 66.1 American Association of Poison Control Centers' Statistics for Exposures of Common Dialyzable Poisons (2021)

Poison	Total Exposures ^a	Deaths
Acetaminophen	137,021	223
Baclofen	4620	11
Barbiturates	1331	3
Carbamazepine	2194	0
Ethylene glycol	6474	20
Gabapentin	20,887	5
Isopropanol	14,706	1
Lithium	6332	10
Metformin	9198	18
(Biguanides)		
Methanol	2228	5
Paraquat	132	1
Phenytoin	1369	2
Salicylates	30,587	26
Theophylline	141	2
Valproic acid	6908	2

^aIncludes coingestions and mixed formulations.

adsorbed by charcoal (e.g., iron or lithium), or illicit drugs in body packers.

Elimination enhancement modalities can be divided between corporeal treatments and ECTRs, which require an artificial device located outside the body.⁴ Expert recommendations endorsed by international toxicology and nephrology societies are available for corporeal treatments^{5,6} and ECTR (<https://www.extrip-workgroup.org/>).⁷⁻²⁶ Physicians must know the physicochemical characteristics, clinical effects, and pharmacokinetics of the poisons to estimate their removal by any technique. This chapter reviews the concepts of poison elimination, elimination enhancement modalities, and poisons requiring the expertise of nephrologists. Fig. 66.1 presents elimination enhancement trends for poisonings in the United States.

OVERVIEW OF CORPOREAL TREATMENTS FOR ENHANCED ELIMINATION OF POISONS

FORCED DIURESIS

The goal of forced diuresis is to induce a urine output that exceeds physiologic flow ($\geq 4\text{--}5$ mL/kg/h) to enhance renal elimination of poisons. Historically, forced diuresis is performed through volume expansion with isotonic fluids (0.9% NaCl or lactated Ringer solution), with or without concomitant diuretics. Results of early studies failed to show significant benefits. Forced diuresis is also associated with complications such as volume overload, pulmonary edema, cerebral edema, and electrolyte disturbances. At present, forced diuresis is never recommended in the management of acute poisonings. However, aggressive volume repletion remains warranted for some poisons to correct hypotension

Table 66.2 Common Antidotes Used in Treatment of Poisonings

Poison	Antidote
Acetaminophen (paracetamol)	N-acetylcysteine
Amanita Phalloides	Silibilin, ^a penicillin G
Anticholinergic drugs	Physostigmine for significant delirium
Anticholinesterase insecticides	Atropine and possibly pralidoxime or obidoxime
Benzodiazepines	Flumazenil
β -adrenergic antagonists	Adrenaline, insulin-dextrose infusion
Calcium channel blockers	Calcium, insulin-dextrose infusion
Carbon monoxide	Oxygen
Cyanide	Hydroxocobalamin and/or thiosulfate
Dabigatran	Idarucizumab
Digoxin	Digoxin Fab antitoxin, atropine
Envenomation (e.g., snake and spider)	Antivenom
Ethylene glycol/methanol	Ethanol or fomepizole
Heparin	Protamine sulfate
Iron	Deferoxamine
Isoniazid	Pyridoxine
Lead	Ca,Na ₂ -EDTA, or succimer (dimercaptosuccinic acid [DMSA])
Mercury	DMSA or DMPS
Methotrexate	Folinic acid, glucarpidase
Nitrates, nitrites	Methylene blue
Opioids	Naloxone
Poison-induced methemoglobinemia (e.g., dapsone, and nitrites)	Methylene blue
Salicylates	Bicarbonate
Sulfonylureas	Octreotide, glucose
Tricyclic antidepressants	Bicarbonate
Valproic acid	L-carnitine
Warfarin	Vitamin K

^aIn the United States, silibilin can be obtained by application for an Emergency Investigational New Drug to the FDA during business hours or the FDA Emergency Call Center after hours.

and/or overcome proximal tubular reabsorption of some offending xenobiotics (e.g., lithium) during volume contraction.

URINARY ALKALINIZATION

Urinary alkalization is based on the concept of "ion trapping." The goal of manipulating urinary pH is to ionize the poison, which decreases passive reabsorption, as ionized particles are less lipid soluble. The ionized poison becomes "trapped" in the tubule lumen and is eliminated in the urine. The dissociation of a weak acid or base into its ionized state is determined by its dissociation constant (pK_a) (i.e., the pH at which it is 50% ionized and 50% nonionized). For example, for salicylic acid (pK_a of 3), when the urinary pH

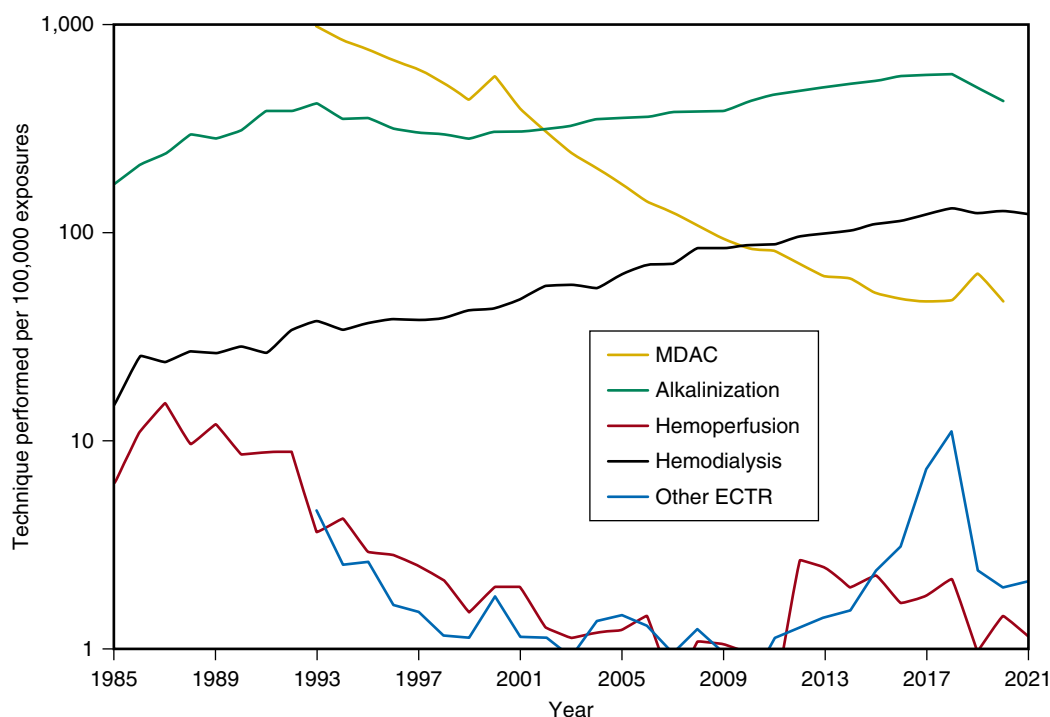


Fig. 66.1 Frequency of various elimination enhancement techniques used in the United States from 1983 to 2021. ECTR, Extracorporeal treatment; MDAC, multiple-dose activated charcoal.

is 3, salicylate exists in a 1:1 ratio of ionized to nonionized forms. At a pH of 7.4, this ratio increases to 20,000:1 and promotes urine excretion.

The clinical efficacy of urine alkalization depends on the relative contribution of kidney clearance to the total body clearance of active poison. If only 1% of the ingested poison is excreted unchanged in the urine, even a 10-fold increase in renal elimination will have no clinically significant effect.⁶ Criteria that determine whether a poison is amenable to urinary alkalization are: 1) it is eliminated unchanged by the kidneys, 2) it is not strongly bound to plasma proteins, 3) it is distributed primarily in the extracellular fluid compartment, and 4) it is a weak acid (i.e., pK_a between 3 and 7).⁶ Urine alkalization enhances the excretion of salicylates, phenobarbital, methotrexate, and other poisons (Table 66.3).⁶

Sodium bicarbonate is administered as an intravenous bolus of 100 to 150 mL of 8.4% sodium bicarbonate in one liter of 5% dextrose in water. This solution is given at up to 200 mL/hour according to the clinical status. Contraindications include severe kidney disease and pulmonary and cerebral edema. Serum electrolytes and urinary pH are monitored every 2 to 4 hours, targeting a urine pH between 7.5 and 8.5 while maintaining a serum pH below 7.55. Complications include hypokalemia, hypocalcemia, hypernatremia, pulmonary and cerebral edema, and metabolic alkalosis. Normokalemia is required for effective urine alkalization; in hypokalemia, potassium is reabsorbed in exchange for hydrogen, so the nephron is unable to produce alkaline urine and the patient can develop alkalemia. Carbonic anhydrase inhibitors (e.g., acetazolamide) are never recommended as they alkalize urine but generate metabolic acidemia, which increases the proportion of the poison unionized in the blood, favoring intracellular distribution and increasing toxicity.

FECAL ELIMINATION ENHANCEMENT

Multiple-dose activated charcoal (MDAC) can enhance elimination of certain poisons. Properties of poisons amenable to MDAC include a low intrinsic clearance, a small volume of distribution (V_D), and a prolonged half-life. MDAC promotes the clearance of poisons by interrupting their enterohepatic circulation, promoting passive diffusion down a concentration gradient from the intestinal capillaries to the intraluminal gut space, and/or limiting the absorption of modified release formulations. Contraindications of MDAC are as those for single-dose activated charcoal (above). The optimal dosage is not confirmed, but it is commonly prescribed as 12.5 g/hour or 50 g every 4 hours, until improvement in clinical status. MDAC is recommended for poisoning due to carbamazepine, dapsone, phenobarbital, quinine, and theophylline⁵ and may also be of benefit in poisonings due to salicylate, colchicine, *Amanita phalloides*, cardiac glycosides,²⁷ or phenytoin (see Table 66.3).²⁸

Ion exchange resins may attract poisons from the gut capillaries to the lumen. Sodium polystyrene sulfonate is used to treat hyperkalemia and can reduce lithium half-life.²⁹ Prussian blue can be used to enhance the elimination of radiocesium and thallium (see Table 66.3).³⁰

PRINCIPLES AND FACTORS INFLUENCING POISON REMOVAL DURING EXTRACORPOREAL TREATMENTS

Elimination of a poison depends on its physicochemical and pharmacologic properties and the ECTR used. Extracorporeal elimination of a poison is significant in these conditions: 1) it can be extracted from the plasma, 2) extracorporeal clearance

contributes to a significant proportion of total clearance, and 3) if the poison exerts its toxicity outside the plasma compartment, a significant proportion of its body stores can be eliminated during ECTR. The first condition depends on the molecular size, water solubility, and protein binding of the poison because these correlate to its extraction ratio (ER) and extracorporeal clearance (CL_{ECTR}). ER can be calculated

as $(A-V)/A$, where A represents the inflow (prefilter) plasma concentration and V represents the outflow (postfilter) plasma concentration. An ER of 1.0 implies the complete removal of a substance from the plasma after a single pass through the extracorporeal circuit. Extracorporeal plasma clearance may be calculated as $CL_{\text{ECTR}} = Q_B \times (1-\text{Hct}) \times \text{ER}$, where Q_B is the blood flow and Hct is the hematocrit. CL_{ECTR} can also be calculated by quantifying the amount of poison in the spent ultrafiltrate and/or dialysate over a period and dividing by its averaged plasma concentration during the same period. The second condition relates to its endogenous clearance, and the third depends on its V_D (see later).

Table 66.3 Poisons Whose Elimination may be Enhanced by Corporeal Techniques

Urine Alkalinization	Multiple-Dose Activated Charcoal	Sodium Polystyrene Sulfonate	Prussian Blue
Chlorophenoxy herbicides (2,4-D, MCPA, MCPP) ^a	Amatoxins (Amanita phalloides)	Lithium Potassium	Radiocesium Thallium
Chlorpropamide	Carbamazepine		
Diflunisal	Colchicine		
Fluoride	Dapsone		
Methotrexate	Digoxin		
Phenobarbital	Phenobarbital		
Salicylates	Phenytoin		
	Quinine		
	Salicylates		
	Theophylline		
	Yellow oleander		

^a2,4-D, 2,4-dichlorophenoxyacetic acid; MCPA, 4-chloro-2-methylphenoxyacetic acid; MCPP, methylchlorophenoxypropionic acid (Mecoprop).

POISON-RELATED FACTORS

MOLECULAR SIZE

All ECTRs will remove poisons with a molecular weight (MW) below 1000 Da, but diffusion and convection will remove them more efficiently. Solutes larger than 1000 Da will be better removed by convection, adsorption, or centrifugation,³¹ although modern intermittent hemodialysis (IHD) can clear poisons up to 10,000 Da (Table 66.4). Convection from hemodiafiltration (HDF) and continuous renal replacement therapy (CRRT) can remove poisons with a MW up to 50,000 Da. Adsorption or centrifugation can remove poisons with a MW exceeding 100,000 Da.

PROTEIN BINDING

The degree of protein binding impacts removal. Hemofiltration and hemodialysis only remove unbound poison, as the poison-protein complex exceeds the pore size of the hemofilter/dialyzer. Diffusion and convection remove poisons

Table 66.4 Summary of Extracorporeal Treatments

	Process	Molecular Weight Cutoff (Da)	Protein Binding Cutoff	Relative Cost	Complications	Comments
Albumin dialysis	Diffusion, adsorption	<60,000-100,000	<95%	++++	++	Liver replacement support
CRRT	Convection and/or diffusion	<10,000-50,000	<80%	++	+	Correction of uremia and acid-base/ E+ disorders
Exchange transfusion	Centrifugation/ Separation, filtration	None	None	++	++	Easier than other ECTRs in neonates
Hemodialysis	Diffusion	<10,000	<80%	+	+	Correction of hemolysis
Hemofiltration	Convection	<50,000	<80%	++	+	Correction of uremia and acid-base/ E+ disorders
Hemoperfusion	Adsorption	<50,000	<95%	++	+++	Saturation of cartridge requires changes
Peritoneal dialysis	Diffusion	<500-5000	<80%	++	++	Low efficacy
Therapeutic plasma exchange	Centrifugation/ separation, filtration	<1,000,000	None	+++	+++	

All extracorporeal treatments above are less likely to be useful for poisons that have a high V_D or a high endogenous clearance. CRRT, Continuous renal replacement therapy; Da, daltons; E+, electrolyte; ECTR, extracorporeal treatment; +, low; ++, moderate; +++, high; +++++, very high.

with a protein binding up to 80%, with a few exceptions. Hemoperfusion can remove poisons with a protein binding up to 90% to 95%, as binding to the adsorbent (activated carbon or, less commonly, a resin) competes with binding to plasma proteins. The clearance depends on the affinity of the poison for the adsorbent.

The degree of protein binding is influenced by the protein concentration, poison concentration, and pathologic states (e.g., uremia).³² In poisonings, saturation of protein binding sites (e.g., valproic acid and salicylate) may occur, increasing the free (unbound) fraction, which promotes removal by ECTR.³³ Phenytoin quickly dissociates from albumin, so the unbound pool is constantly being removed by ECTR.

ENDOGENOUS CLEARANCE

Extracorporeal removal will be useless if the endogenous clearance of a poison far outweighs its ECTR clearance.³⁴ This explains why hemodialysis is futile for cocaine or toluene, which have high endogenous clearances (>4 mL/minute/kg). Additionally, significant impairment in endogenous metabolic and/or elimination pathways may increase the efficacy of ECTRs; for example, the renal clearance of metformin decreases from 500 mL/min to 0 mL/min in anuria, rendering ECTR clearance clinically significant.

VOLUME OF DISTRIBUTION (V_D)

The V_D is the apparent volume where a poison is distributed at equilibrium before clearance occurs (e.g., with rapid intravenous administration) and is calculated by dividing the total poison in the body by its concentration. This relationship assumes a single homogenous compartment of water in which the poison is distributed.

Poisons that distribute extensively in tissues (e.g., tricyclic antidepressants) have a high V_D ; conversely, poisons that distribute in total body water, such as methanol, have a low V_D (<1 L/Kg). Because ECTRs only remove poisons in the intravascular space, they will not significantly eliminate poisons with a high V_D . For example, digoxin easily crosses the dialyzer; however, because of its high V_D (7 L/kg), <5% of the total body burden will be removed during a 6-hour IHD. Many publications erroneously conclude that poisons with a high V_D are amenable to ECTR removal on the basis of a high extracorporeal clearance or reduction of serum concentrations.³⁵⁻³⁷ ECTR may occasionally be considered for poisonings with a high V_D (e.g., methotrexate and thallium) early after exposure and before distribution to tissues, when there is still a high burden of poison in the blood.⁷

AVAILABLE EXTRACORPOREAL TREATMENTS TO ENHANCE ELIMINATION OF POISONS

Techniques facilitating poison removal are classified by their mechanism of removal: diffusion (hemodialysis, peritoneal dialysis [PD]), convection (hemofiltration), adsorption (hemoperfusion with charcoal, CytoSorb), and centrifugation (therapeutic plasma exchange [TPE]).^{38,39} The features of these treatments are summarized in Table 66.4.

INTERMITTENT HEMODIALYSIS

During IHD, the poison diffuses from the blood compartment to a dialysate flowing in a countercurrent direction, separated

by a semipermeable dialysis membrane. The principles that dictate solute removal in IHD also apply to poison elimination (see Chapter 62). The characteristics of a xenobiotic that favor efficient removal by IHD are low endogenous clearance (<4 mL/min/kg), low V_D (<1–2 L/kg), low MW (<5000–10,000 Da), and low protein binding (<80%).

The type of dialysis membrane, its surface area, pore size, and the blood and dialysate flow rates will influence poison clearance (Table 66.5). New high-flux membranes and catheters allow the removal of larger-sized poisons considered nondialyzable 30 years ago.³¹ Increasing the dialysate flow rate and, more importantly, the blood flow rate will promote diffusion and elimination of the poison.³⁹ Filters with large surfaces and maximal blood and dialysate flow rates should be used unless contraindicated (e.g., concern for disequilibrium syndrome).³⁹

IHD has several advantages over other ECTRs in poisonings because it rapidly removes poisons and corrects volume overload, acid-base, and electrolyte abnormalities.⁴⁰ IHD is also more available, less expensive, and quicker to implement.⁴¹

The most common acute complication of IHD is hypotension, which is seen in patients with acute kidney injury (AKI) or end-stage kidney failure (ESKF) requiring fluid removal. Because poisoned patients rarely require net ultrafiltration, dialytic hypotension more likely reflects the effect of the poison. While hypotension is often a marker of poisoning severity and can be an indication for ECTR, it may limit the ECTR's performance.

HEMOPERFUSION

During hemoperfusion, blood circulates through an extracorporeal circuit equipped with a charcoal or resin cartridge onto which the poison is adsorbed.⁴² Compared with diffusion, adsorption is not as limited regarding the MW, lipid solubility, or protein binding of the poison. Alcohols and most metals, however, are poorly adsorbed. Although certain exchange resins (XAD-4) can remove organic solutes and nonpolar (lipid soluble) poisons, they are unavailable in some countries (e.g., the United States).

Current hemoperfusion devices have improved biocompatibility. Coating of the sorbent material minimizes direct contact between the adsorbent and blood constituents, reducing the risk of embolization without impairing its adsorptive capacity. The circuit requires more anticoagulation than IHD and blood flows are limited to 350 mL/min because of hemolysis.⁴³

Table 66.5 Factors that May Enhance Poison Clearance During Hemodialysis

Larger surface area of dialysis membrane
High-flux dialyzer
High blood and dialysate flows
Increased ultrafiltration rate (with replacement solution)
Increased time on dialysis
Reduced recirculation
Two dialyzers in series
Two distinct extracorporeal circuits

Hemoperfusion carries a higher risk of complications than IHD,⁴⁴ namely the nonselective adsorption of certain cells and molecules.^{45,46} Thrombocytopenia, hypoglycemia, and hypocalcemia can occur within hours.⁴⁴ Cartridge costs are several-fold higher than for dialyzers and need to be replaced every 3 to 4 hours due to saturation. With technologic advancements in IHD, clearances of theophylline and phenobarbital by hemoperfusion or IHD are now comparable.^{42,44,47} Hemoperfusion does not correct electrolyte and acid-base disturbances or fluid overload and is less available. Consequently, IHD is preferred in most settings where hemoperfusion is indicated.⁴²

Recently, CytoSorb, a hemoabsorption device containing polymer beads integrated into CRRT prefilter, has been developed to remove cytokines in septic shock. The device can adsorb xenobiotics with a MW up to 60 kDa and must be changed after 4 to 8 hours.⁴⁸ Although there is experience with poisonings, it is uncertain if CytoSorb provides any advantage over IHD, given that most poisons are small. There are reports of successful treatment of amitriptyline, quetiapine, venlafaxine, digitoxin, and flecainide poisonings.⁴⁸⁻⁵⁰ However, given the large V_D of these poisons, the impact of CytoSorb on clinical outcomes is uncertain.

HEMOFILTRATION

During hemofiltration (HF), solute and solvent are removed by solvent drag (convection) and replaced by a physiologic solution. Clearance is dependent on the sieving coefficient (SC) of the membrane, which is the ratio of a solute concentration in the ultrafiltrate to its concentration in the plasma. A solute that freely crosses the membrane has an SC of 1, while a SC of 0 means that the solute cannot cross the membrane. Poison elimination by HF depends on factors similar to those regulating diffusion. However, convective transport also allows the removal of larger poisons, up to 50,000 Da,³⁹ which has limited relevance in toxicology because most poisons have a MW below 2000 Da.

COMBINED THERAPIES

Diffusion and convection are increasingly combined (hemodiafiltration). Adsorption (hemoperfusion) and diffusion (IHD) are sometimes used concurrently, in series, to maximize poison clearance. However, the clearance of this combination is not additive and increases complications. Clearances are additive if the therapies are implemented in parallel, with two distinct circuits.

CONTINUOUS RENAL REPLACEMENT THERAPY

Most ECTRs can be offered intermittently or continuously. CRRT is popular in the intensive care unit (ICU) to manage AKI, and available in various forms (see Chapter 64): continuous venovenous hemodialysis (CVVHD), continuous venovenous hemofiltration (CVVH), and continuous venovenous hemodiafiltration. Their role in poisoning remains uncertain.⁵¹ The solute clearance with CRRT is lower than with IHD because of lower blood and effluent flow rates. In AKI, CRRT is often chosen due to better tolerability for fluid removal in shock. However, in poisoned patients, net ultrafiltration is rarely required and IHD therefore preferred given its superior poison clearance.

CRRT may be used after IHD to avoid a rebound in poison concentration; however, the benefit of this practice is uncertain (see later).

Sustained low-efficiency dialysis (SLED) is a hybrid technique between IHD and CRRT,^{52,53} which provides a lower poison clearance than IHD over the same period.⁵⁴

PERITONEAL DIALYSIS

PD clears substances by diffusion but has a limited role in acute poisoning, as the maximum clearance is 15 mL/min, representing <10% of that achieved during IHD. However, PD does not require an extracorporeal circuit and may be easier to perform in neonates and the only technique available in resource-poor settings.

THERAPEUTIC PLASMA EXCHANGE AND PLASMAPHERESIS

Plasmapheresis involves the withdrawal of venous blood and separation of plasma from blood cells by filtration or centrifugation and reinfusion of cells with autologous plasma or another replacement solution (Chapter 65). In TPE, the removed plasma is discarded and replaced by 5% albumin or fresh-frozen plasma. Clearance is limited by the plasma removal rate (≈ 50 mL/min).^{38,55} TPE is considered for highly protein-bound (>95%) or large poisons (>50,000 Da), such as cisplatin⁵⁶, L-thyroxine⁵⁷, and vincristine.⁵⁸ TPE has been used to treat near-fatal infusion reactions to rituximab.⁵⁹ Complications include those associated with catheter installation, bleeding, hypocalcemia, and hypersensitivity reactions to the replacement solution.^{60,61}

EXCHANGE TRANSFUSION

During exchange transfusion (ET), whole blood or red blood cells are removed by apheresis and replaced with transfused blood products. ET may be considered in poisons causing massive hemolysis (e.g., sodium chlorate or nitrite poisoning in a patient with G6PD deficiency) or in infants, as it is technically simpler than IHD.

EXTRACORPOREAL LIVER ASSIST DEVICES (ALBUMIN DIALYSIS)

Albumin dialysis is historically used to replace liver function in fulminant hepatitis or severe cirrhosis. There are three types of "albumin dialysis." Single-pass albumin dialysis (SPAD) is similar to CRRT with albumin added to the dialysate. Molecular Adsorbents Recirculation System (MARS) is identical to SPAD, but the albumin-enhanced dialysate (with the adsorbed poison) is recycled after going through a dialysis filter with resin and charcoal cartridges. Prometheus combines albumin adsorption with IHD after selective filtration of the albumin fraction through a polysulfone filter.

These devices can theoretically remove albumin-bound xenobiotics, bile acids, and bilirubin better than IHD or CRRT and have been used to support liver function in hepatotoxicity caused by acetaminophen and amatoxin-containing mushroom poisonings.⁶²⁻⁶⁵ However, they provide inferior clearances for theophylline, valproic acid, and phenytoin.⁶⁶⁻⁶⁸ Their role remains unclear due to their limited clearance and availability, as well as high cost and complication rates.

GENERAL INDICATIONS FOR EXTRACORPOREAL REMOVAL OF POISONS

The EXTRIP (EXtracorporeal Treatment In Poisoning, <http://www.extrip-workgroup.org/>) workgroup^{69,70} has completed guidelines for the use of ECTR for several poisonings^{7-18,71} and proposed directions for future research.⁷⁰ The decision to initiate any form of blood purification must take into account a risk assessment, weighing benefits and risks/costs expected from extracorporeal treatments and poison-related factors (Fig. 66.2). Absolute Indications for ECTR Include the Following (All Must Be Present).⁴⁰

1. **Severe toxicity from the exposure (either current or expected).** The exposure to a specific poison must be significant enough to warrant active poison removal with ECTR. A proper risk assessment of the exposure, which includes close collaboration with a poison control center, may help to estimate the risk for a specific patient. In some cases, an estimation of ingestion, expressed in dose/weight (e.g., mg/kg), or blood concentrations might predict future clinical compromise and prompt transfer to a dialysis center and *prophylactic* ECTR (i.e., before the appearance of severe toxicity).

2. **There are no preferable alternative treatments.** Antidotes may either amend or prevent the apparition of toxicity related to a poison and are usually preferred to ECTR, as they are less invasive. For example, this is the case for most acetaminophen poisonings, when N-acetylcysteine is available.

3. **The ECTR must be able to remove the poison** (see earlier).

The clinician must anticipate which benefits are expected from the procedure; in some exposures, ECTR is life-saving (e.g., high-dose salicylate and paraquat) or prevent irreversible tissue damage (e.g., blindness from methanol). The advantages of ECTR in those circumstances would largely outweigh the costs and complications of the procedure. In other situations, ECTR may reduce the length of coma and mechanical ventilation, as seen in poisonings causing central nervous system (CNS) depression (e.g., barbiturates and anticonvulsants). Finally, ECTR may reduce hospital length of stay and associated costs from antidotes (e.g., dialysis versus fomepizole alone in a methanol-poisoned patient without metabolic acidosis).⁷²

These benefits must be weighed against: 1) mechanical risks of ECTR including those associated with catheter insertion (which can be minimized with ultrasonographic guidance)⁷³; 2) removal of certain antidotes (e.g., ethanol and N-acetylcysteine)⁷⁴; 3) withdrawal syndromes from

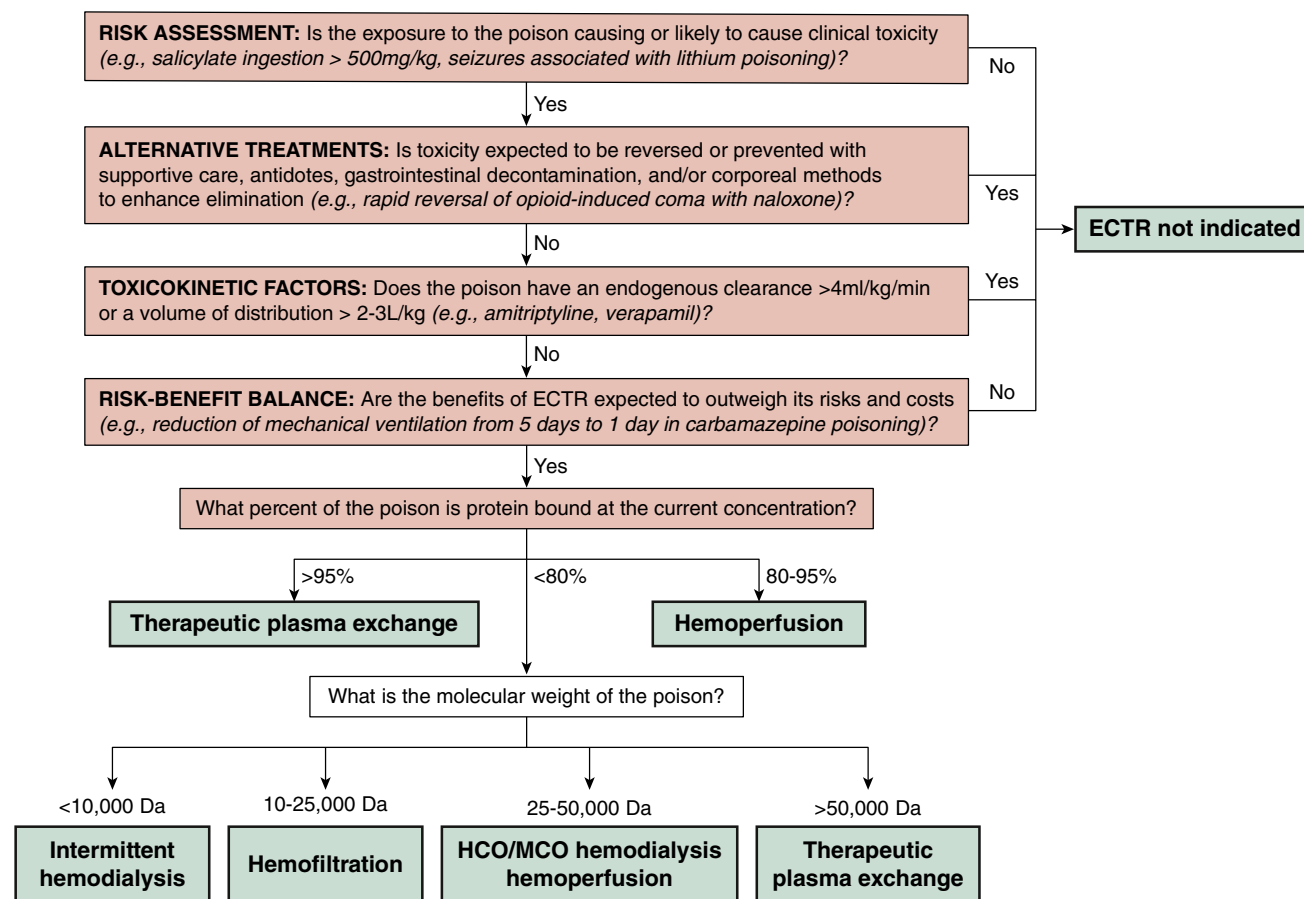


Fig. 66.2 Practical approach for using an extracorporeal treatment in poisonings. This assessment requires analysis of the patient's condition, the exposure, alternative treatments, molecular and toxicokinetic characteristics, and available modalities. ECTR, Extracorporeal treatment; HCO, high cutoff; MCO, medium cutoff; V_D , volume of distribution.

a rapid decrease in drug concentration in patients with physiologic tolerance to a xenobiotic from chronic use; and 4) cost of a single dialysis including equipment and nursing/physician fees, which are usually minor compared with the cost of a day in the ICU.⁴¹ In the absence of any clinical outcome data, studies should demonstrate, at a minimum, significant poison removal.

TECHNICAL CONSIDERATIONS

Patients with poisonings are different from patients with AKI or ESKF, and the prescription of any ECTR should reflect these differences.

Vascular access: A double-lumen central catheter is required for most ECTRs. Because time is important, a temporary catheter is preferred, using ultrasound guidance to reduce complications.⁷⁵ Inserting a femoral line does not require x-ray assistance but increases recirculation compared with subclavian or right jugular sites.⁷⁶ A femoral access creates less malfunction than a left jugular catheter.⁷⁷ Dual catheter sites and circuits may maximize poison clearance.^{78,79}

Choice of dialyzer/filter/adsorber: For IHD, high-flux dialyzers with the largest surface area should be used. The dialyzer or hemofilter should have a molecular size cut-off above that of the poison. For hemoperfusion, the only column licensed in the United States and Canada is the Gambro Adsorba 300c, a charcoal cartridge.⁴²

Anticoagulation: Unfractionated or low-molecular-weight heparin prevents clotting and maintaining the patency of the circuit. In patients at high risk of bleeding, heparin is substituted by saline flushes. For hemoperfusion, heparin is required in greater quantities than IHD to reduce the risk of hemolysis.⁴³

Blood, dialysate, and effluent flows: Should be maximized to increase clearance according to the manufacturer.³⁹

Dialysate composition: The sodium, bicarbonate, potassium, calcium, and magnesium concentrations in the dialysate or replacement fluid (RF) are adjusted according to the serum concentrations. Phosphate may be added to the dialysate or RF to avoid hypophosphatemia. Periodic measurements of serum biochemistry should be performed, and the dialysate modified accordingly.

Duration of ECTR: A 6-hour high-efficiency ECTR is usually sufficient. When significant toxicity is present or may be prolonged, or if a less efficient therapy (CRRT, SLED) is used, treatment can be extended. Hemoperfusion cartridges should be replaced after 3 to 4 hours due to saturation.

Patient disposition: Many poisoned patients die before dialysis.⁸⁰ If the risk assessment for a suspected toxic exposure suggests that dialysis may be required, even if the patient does not yet meet the criteria, preemptive transfer to a center that can provide ECTR is justified. Because significant delay may occur before ECTR initiation,⁴¹ a temporary dialysis catheter should be inserted and a transfer to the ICU expedited.⁴⁰

Rebound: Rebound is a sudden increase in poison concentration after the cessation of ECTR, which typically occurs after IHD/hemoperfusion. Rebound may be due

to 1. redistribution of poison from deep compartments (tissues and intracellular space) to the plasma (lithium,⁸¹ dabigatran,⁸² and methotrexate⁸³), especially in poisons with a large V_D ; or 2. ongoing absorption. In 1., the rebound might reflect a decrease in the poison concentration from the toxic compartments (lithium), which may be beneficial.⁸⁴ In 2., the rebound can cause recurrent symptoms. If the rebound is of concern, a clinician may choose to repeat or extend ECTR or perform continuous therapy.⁸⁵ Following ECTR, serial poison concentrations and clinical status should be monitored to identify redistribution (≈ 12 –24 hours) or ongoing absorption.

POISONS AMENABLE TO EXTRACORPOREAL ELIMINATION

In most poisoning cases, ECTRs are not required. Poisons commonly responsible for fatalities (e.g., tricyclic antidepressants, short-acting barbiturates, stimulants, and illicit street drugs) are not effectively amenable to ECTR.⁸⁶ The remainder of this chapter focuses on the characteristics of poisons for which ECTR may be considered. Physicochemical characteristics of major poisons are shown in Table 66.6, and common indications for ECTR in poisoning in Fig. 66.3.

TOXIC ALCOHOLS: ETHYLENE GLYCOL, METHANOL, AND ISOPROPANOL

Ethylene glycol (EG) is found in antifreeze, radiator fluid, solvents, hydraulic brake fluid, deicing solutions, and polishes. Methanol (wood alcohol) is found in windshield or glass cleaning solutions, printing solutions, dyes, thinners, fuels, and antifreeze additives for gasoline. Isopropanol (i.e., isopropyl alcohol or 2-propanolol) is found in “rubbing alcohol,” skin lotion, aftershave lotion, denatured alcohol, solvents, and cleaning products.

TOXICOLOGY AND TOXICOKINETICS

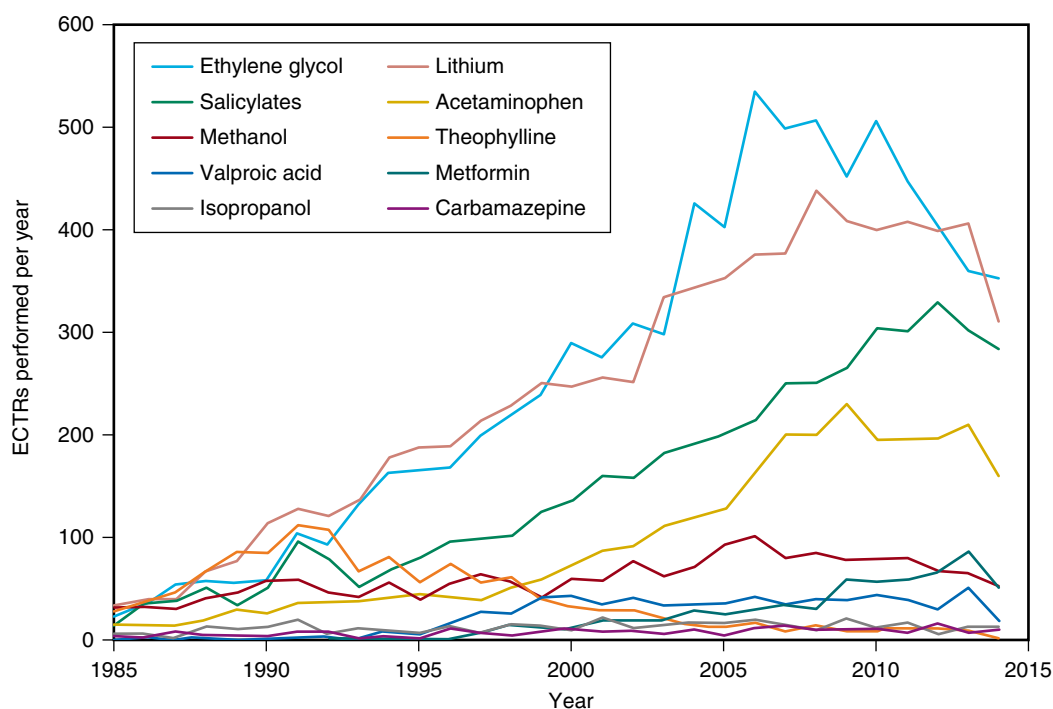
EG, methanol, and isopropanol are small unbound molecules with a small V_D (0.6 L/Kg). Peak concentrations occur rapidly after ingestion. Approximately 30% of EL is excreted unmetabolized by the kidneys; the remainder is oxidized to glycolaldehyde by alcohol dehydrogenase (ALDH) in the liver and converted to glycolic acid by aldehyde dehydrogenase, followed by the conversion of glycolic acid to glyoxylic acid (Fig. 66.4, A). Metabolic acidosis results mainly from the accumulation of glycolic acid,⁸⁷ while oxalate precipitates in tissues including kidneys, causing AKI. Glycolate cross-reacts with some lactate assays on point-of-care blood analyzers, which falsely elevates serum lactate levels. Any EL ingestion over 0.1 mL/kg requires treatment. Deaths have been reported with ingestions as low as 30 mL.

Methanol is metabolized principally by ALDH (85%) to formaldehyde and then oxidized by formaldehyde dehydrogenase to formic acid (see Fig. 66.4, B). In a folate-dependent step, formic acid is transformed to water and CO_2 . Formic acid inhibits cytochrome c oxidase in cells’ mitochondria and is responsible for toxic symptoms.⁸⁸ The lethal dose of methanol is 10 mL.⁸⁹

Table 66.6 Physiochemical Characteristics and Toxicokinetics of Various Poisons

	Molecular Weight (Da)	Protein Binding	Volume of Distribution (L/Kg)	Endogenous Clearance in Healthy Adults (mL/min/kg)
Acetaminophen	151	20%	1	5
Atenolol	266	5%	1.1	2.1
Baclofen	214	30%	0.9	3
Carbamazepine	236	75%	1.2	1.3
Ethylene glycol	62	0%	0.6	1.8
Gabapentin	171	0%	0.9	1.5
Isoniazid	137	10%	0.7	2 (slow acetylator) 7 (rapid acetylator)
Isopropanol	60	0%	0.6	1.2
Lithium	7	0%	0.8	0.4
Metformin	166	5%	5	10
Methanol	32	0%	0.6	0.7
Methotrexate	454	50%	0.8	1.5
Paraquat	186	5%	1.0	8
Phenobarbital	232	40%	0.7	0.2
Phenytoin	252	90%	0.6	0.4
Pregabalin	159	0%	0.5	1.1
Salicylic acid	138	80% ^a	0.2	1.5
Sotalol	272	0%	1.4	2
Theophylline	180	60%	0.5	0.7
Valproic acid	144	90% ^a	0.2	0.1

^aProtein binding saturation occurs at high concentrations.
Da, daltons.

**Fig. 66.3** Frequency of extracorporeal treatments used in the United States from 1983 to 2015. *ECTR*, Extracorporeal treatment.

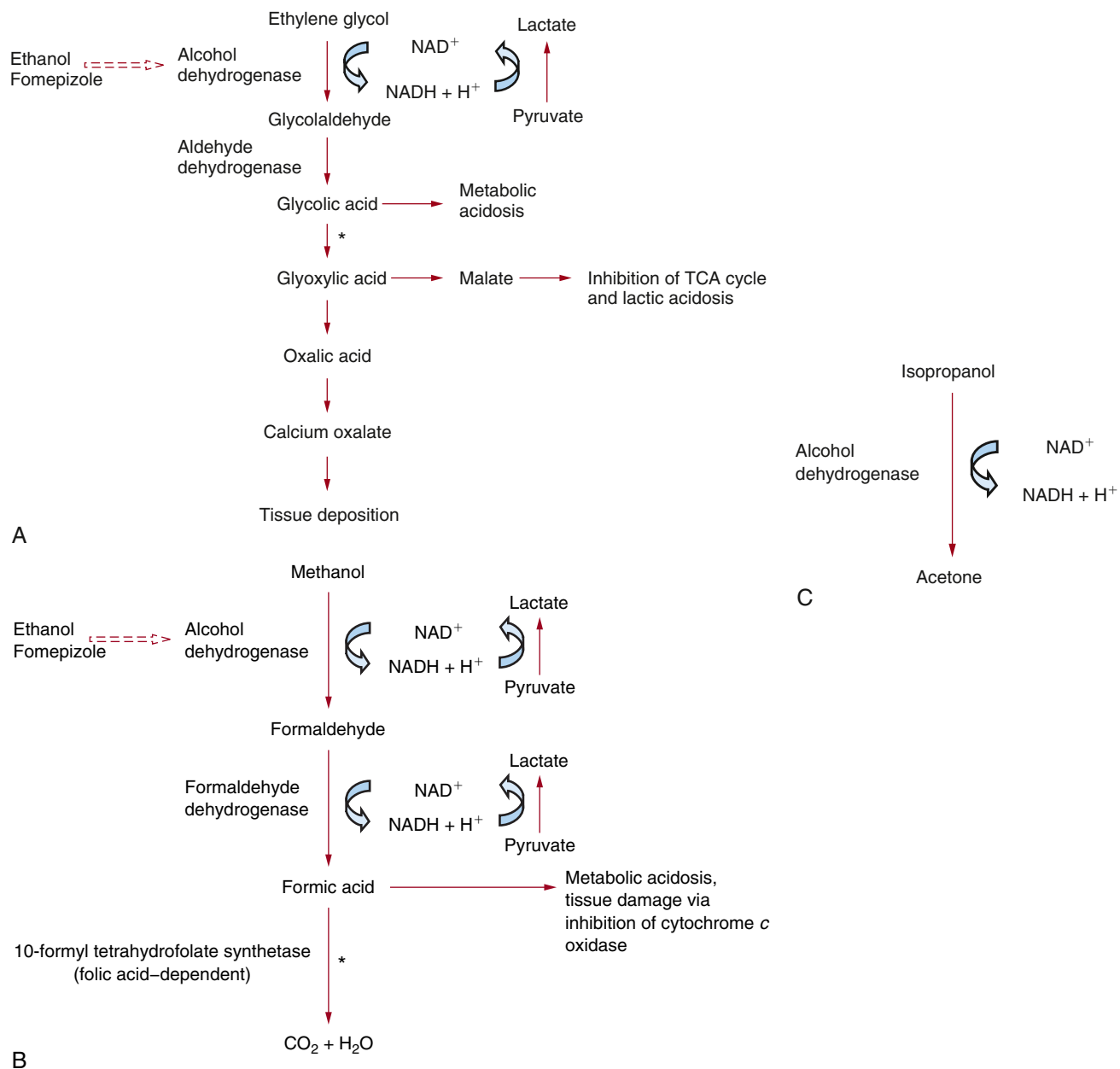


Fig. 66.4 Metabolism of toxic alcohols. (A) Ethylene glycol: The *broken arrow* points to inhibitors of alcohol dehydrogenase; the *asterisk* denotes the rate-limiting step. In the presence of the electron acceptor, nicotinamide adenine dinucleotide (NAD^+), ethylene glycol is oxidized to glycolaldehyde by alcohol dehydrogenase. Aldehyde dehydrogenase then rapidly converts glycolaldehyde to glycolic acid, which is followed by the slow conversion of glycolic acid to glyoxylic acid (the rate-limiting step). Final end products include oxalic acid, glycine, and oxalomalic acid, which are all effectively removed by intermittent hemodialysis (*IHD*). (B) Methanol: The *broken arrow* points to inhibitors of alcohol dehydrogenase; the *asterisk* denotes the rate-limiting step. (C) Isopropanol. $NADH + H^+$, reduced form of nicotinamide adenine dinucleotide; *TCA*, tricarboxylic acid.

Approximately 80% of isopropanol is metabolized to acetone via ALDH, and the remainder is eliminated unchanged in urine (Fig. 66.4, C).⁹⁰ Isopropanol displays first-order elimination kinetics with an elimination half-life of 3 to 8 hours,^{91,92} while the half-life of acetone is 10 hours.^{93,94} Most of the CNS depressant effects are attributed to isopropanol.⁹¹ The lethal dose is 150 to 250 mL.⁹⁵

CLINICAL PRESENTATION

All toxic alcohols initially cause inebriation, similar to that seen with ethanol. As EL is metabolized, metabolic acidemia

appears after a latent period of approximately 3 to 6 hours after ingestion, with progressive neurotoxicity (coma, seizures, cerebral edema, cranial nerve defects, radiculopathies, brainstem injury); cardiotoxicity (hypotension); and respiratory distress. AKI occurs in approximately 50% of patients due to the precipitation of calcium oxalate crystals in the kidneys and direct tubular injury from glycolate. If left untreated, EL poisoning leads to multiorgan failure and death in up to 20% of patients.

Methanol poisoning results in neurologic, visual, and gastrointestinal (GI) symptoms. CNS manifestations may

include inebriation, headaches, dizziness, nausea, seizures, and cerebral edema. Methanol can produce a Parkinson-like syndrome through damage to the putamen and subcortical white matter of the basal ganglia.^{96,97} Metabolic acidosis, caused by accumulation of formate and lactate, can be severe. Vision deficits (blurred vision and decreased visual acuity) are the hallmark of methanol poisoning and usually occur 6 to 30 hours after exposure, depending on how much ethanol is coingested. Vision deficits are permanent in 25% of cases. Death is usually caused by cardiovascular shock and respiratory arrest.

Isopropanol overdose is characterized by altered sensorium, which may range from mild inebriation, lethargy, stupor, respiratory depression, and/or coma. Isopropanol irritates the GI tract causing nausea, vomiting, gastritis, and abdominal pain. Isopropanol is toxic to myocytes and can induce severe reversible hypotension.⁸⁹ Other findings include hypoglycemia from impaired gluconeogenesis, hypothermia, and hemolytic anemia.

DIAGNOSTIC TESTING

Toxic alcohol assays are not widely available.⁹⁸ EG and methanol poisoning should be considered in any patient presenting with a high anion gap metabolic acidosis and/or a high osmol gap.⁹⁹ In early presentations, the osmol gap is more prominent, whereas in later presentations, a high anion gap predominates. As toxic alcohols are osmotically active, the osmol gap, calculated as the difference between measured osmolality (by freezing point depression) and calculated osmolality, approximates the toxic alcohol concentration (in mmol/L) and can be converted to mg/dL (Table 66.7). The osmol gap can be monitored serially during admission and dialysis when serum concentrations are unavailable.^{100,101} The calculated osmolality is based on the concentrations of sodium, glucose, blood urea nitrogen, and ethanol:

$$\begin{aligned} \text{Toxic alcohol concentration (in mmol/L)} &\cong \text{Osmol gap} \\ &= \text{Measured Osmolality} - \text{Calculated Osmolality} \\ &= \text{Measured Osmolality} - (2 \times \text{Na} + \text{glucose} + \text{urea} + 1.2 \times \text{ethanol}) \\ &\quad \text{SI units} \\ &= \text{Measured Osmolality} - (2 \times \text{Na} + \text{glucose}/18 + \text{BUN}/2.8 + \text{ethanol}/4) \\ &\quad \text{Imperial units} \end{aligned}$$

Table 66.7 Conversion of Toxic Alcohol Concentration From mmol/L to mg/dL

Toxic Alcohol	Conversion from mmol/L to mg/dL
Acetone	x 5.81
Diethylene glycol	x 10.61
Ethanol	x 4.61
Ethylene glycol	x 6.21
Isopropanol	x 6.01
Methanol	x 3.20
Propylene glycol	x 7.61

The normal osmol gap is between -5 and 10. In isopropanol poisoning, isopropanol and acetone contribute to the osmol gap.¹⁰² A “normal” osmol gap does not exclude the diagnosis of toxic alcohol poisoning: If a specific patient has a baseline gap of 0, ingestion of 25.6 mg/dL or 8 mmol/L of methanol would still lead to a “normal” osmol gap. Also, the osmol gap remains normal if a patient presents late after ingestion, once the parent alcohol has undergone complete oxidation since glycolic acid and formic acid do not contribute to the osmol gap.¹⁰³

Similarly, the anion gap can be used to estimate the glycolic acid and formic acid concentrations. The anion gap is the difference between measured cations (Na^+ and K^+) and anions (Cl^- and HCO_3^-), representing the difference between unmeasured anions and cations, in mmol/L. Lactate can also contribute to the anion gap and needs to be factored in the equation:

$$\begin{aligned} \text{Glycolic acid/Formic acid concentration} \\ = \text{Na} + \text{K} - \text{Cl} - \text{HCO}_3 - \text{Lactate} - 14 \end{aligned}$$

Studies have shown a good correlation between the concentration of metabolites and the anion gap.^{104,105} The absence of an anion gap suggests early presentation after EG or methanol exposure before metabolism, coingestion of ethanol, or another alcohol (isopropanol and propylene glycol).

Calcium oxalate crystals are found in the urine sediment of 50% of patients with EG exposure, 4 to 8 hours after ingestion,^{106,107} but can also be seen with large ingestions of vitamin C or high oxalate-containing food. Urine that fluoresces under Wood lamp illumination is a unique feature of EG poisoning attributed to many types of antifreeze, which can be detected for up to 6 hours after ingestion.^{108,109} Other laboratory abnormalities are hypocalcemia and leukocytosis.

Characteristic findings of isopropanol exposure include increased osmol gap, absent or low-grade metabolic acidosis with a normal anion gap, fruity acetone breath, acetonemia, and acetonuria without hyperglycemia. Acetonemia or acetonuria can be suspected by a positive sodium nitroprusside reaction in plasma or urine. A low concentration of serum ketones 2 hours after isopropanol ingestion (in absence of ALDH inhibition) generally excludes substantial ingestion.⁹¹

TREATMENT

In toxic alcohol poisonings, clinicians often must initiate treatment before confirming the exposure with specific laboratory tests. Management includes supportive care, correction of acidemia, antidotal therapy, and ECTRs. High-minute ventilation should be maintained if significant acidemia is present. Because toxic alcohols are rapidly absorbed from the GI tract, decontamination is seldom performed. Aspiration of the gastric contents may be useful if the ingestion is recent.

The cornerstone of treatment for EG or methanol poisoning is to prevent their metabolism into toxic metabolites using an antidote (i.e., ethanol or fomepizole, which competes for ALDH).

Indications for these antidotes are^{99,110}:

1. Serum concentration of EG >20 mg/dL (3.2 mmol/L) or methanol >20 mg/dL (6.3 mmol/L)
2. Documented recent (hours) ingestion of toxic amounts of EG or methanol and osmol gap >10
3. A history or strong clinical suspicion of EG or methanol poisoning and at least two of the following: arterial pH <7.3, serum bicarbonate <20 mEq/L, osmol gap >10, or oxalate crystals in urine

ALDH inhibitors are contraindicated in isopropanol poisoning; the metabolite (acetone) has limited toxicity, and administration of ethanol and fomepizole can prolong the CNS depressant effect of isopropanol.¹¹¹ Ethanol can be given enterally, intravenously, or via the dialysate. The intravenous formulation allows immediate bioavailability and avoids GI distress. Target serum ethanol concentrations for ALDH competition are between 100 and 150 mg/dL (21.7–32.5 mmol/L); below this concentration, ALDH inhibition would not maximally inhibit metabolite formation. Above this range, there is a high risk of respiratory depression.¹¹⁰ Fomepizole (4-methylpyrazole, Antizol) has largely replaced ethanol as the primary therapy and has multiple advantages over ethanol: 1) more potent ALDH inhibition, 2) simple dosing, 3) predictable pharmacokinetics, 4) no blood monitoring, 5) few side effects and no CNS depression, and 6) longer duration of action. Its main drawback is the cost (US\$1000 per dose). Either antidote is usually continued until EG or methanol concentrations are below 20 mg/dL (EG <3.2 mmol/L or methanol <6.3 mmol/L) and the patient is asymptomatic with a normal arterial pH. Table 66.8 presents dosages of ethanol and fomepizole during toxic alcohol poisonings.

Acidemia should be corrected with the administration of IV sodium bicarbonate,¹¹² which enhances deprotonation of acid metabolites, making them less likely to penetrate end-organ tissues (retina and kidney) and increasing their excretion by the kidneys. An intravenous bolus (1–2 mEq/kg), followed by an infusion, if necessary, should be given to maintain an arterial pH between 7.45 and 7.55. Asymptomatic hypocalcemia is not routinely treated in EG poisoning because it could exacerbate calcium oxalate crystal formation and deposition. Pyridoxine, thiamine, and magnesium are cofactors in the metabolism of EG, and their supplementation is recommended in malnourished patients or those with known deficits.⁹⁹ Folic acid supplementation

(50 mg IV every 4 to 6 hours for 5 doses) is recommended for methanol poisoning.

ECTRs, especially IHD, are extremely efficient at removing alcohols and their toxic metabolites, as well as correcting metabolic acidosis. With optimal dialysis parameters, clearance of alcohols and metabolites can reach 250 mL/min.

The increasing availability of fomepizole has modified the indications and pertinence of ECTR because of its great efficacy in preventing metabolite formation.¹¹³ Patients with EG poisoning without acidosis or kidney impairment may be treated with fomepizole alone, whatever the concentration of EG.¹¹⁴ The same applies to methanol; however, endogenous clearance of methanol with fomepizole is low. Assuming a methanol half-life of 54 hours with fomepizole,¹¹⁵ a patient with a methanol concentration of 320 mg/dL (100 mmol/L) would need 9 days to reach a safe methanol concentration (<20 mg/dL or 6.3 mmol/L). Dialysis might reduce hospitalization costs and antidote requirements for patients poisoned with EG or methanol.^{72,116} IHD is mandatory with metabolic acidosis or an increased anion gap because this indicates toxic metabolite accumulation.

Indications for ECTR for EL and methanol poisoning include the following^{15,19}:

- Serum methanol concentration >50 mg/dL (15.6 mmol/L) or serum EG >50 mmol/L
- Osmol gap >50 due to methanol or EG
- Serum glycolate concentration is >8–12 mmol/L
- Anion gap calculated with K ≥24 mmol/L
- Metabolic acidosis (pH <7.2)
- Coma or seizures
- Vision deficits due to methanol
- AKI or CKD

The expected duration of dialysis (hours) can be estimated using an estimated half-life of EG or methanol of 2.5h during IHD^{117,118} provided that metabolic acidosis is corrected. Monitoring of the toxic alcohol concentration or osmol gap is recommended to confirm the estimation.¹¹⁹ IHD should be continued until the parent alcohols are <20 mg/dL (EG <3 mmol/L or methanol <6 mmol/L) and metabolic acidosis is corrected. CRRT offers lower clearance and requires longer treatment but may be considered if IHD is unavailable.^{15,19}

IHD effectively removes isopropanol and acetone and is indicated when the isopropanol concentration is >400 mg/

Table 66.8 Antidote Dosage During Toxic Alcohol Poisoning

	40% PO Ethanol (e.g., scotch)	10% IV Ethanol	IV Fomepizole
Loading dose ^a	1.8 mL/kg	7.6 mL/kg	15 mg/kg
Maintenance dose	0.2 mL/kg/h (nondrinker) 0.5 mL/kg/h (chronic drinker)	0.8 mL/kg/h (nondrinker) 2.0 mL/kg/h (chronic drinker)	10 mg/kg q12h x 4 doses, then 15 mg/kg q 12 hours
Maintenance dose during IHD	0.5 mL/kg/h (nondrinker) 0.8 mL/kg/h (chronic drinker)	2.1 mL/kg/h (nondrinker) 3.3 mL/kg/h (chronic drinker)	Same dose but q4h or a constant infusion of 1.0–1.5 mg/kg/h

^aAssumes initial ethanol concentration is zero, dose is independent of chronic drinking status.

dL (66.6 mmol/L), or with prolonged coma, hypotension, myocardial depression, tachydysrhythmias, or AKI.¹²⁰⁻¹²²

The use of heparin should preferably be avoided in methanol-poisoned patients due to the increased risk of intracerebral hemorrhage.

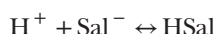
SALICYLIC ACID

Salicylates are analgesics, antipyretics, and antiinflammatories, and aspirin (acetylsalicylic acid) is commonly prescribed for its antiplatelet properties. Topical salicylic acid is a keratolytic and wart remover, oral bismuth subsalicylate (Pepto-Bismol) treats gastroesophageal reflux, and topical methyl salicylate is an analgesic and antiinflammatory.^{123,124}

TOXICOLOGY AND TOXICOKINETICS

Salicylates are rapidly absorbed from the GI tract. Peak serum concentrations occur within an hour, except for enteric-coated products. In acute overdose, bezoar formation and pylorospasm may prolong absorption and the onset of symptoms can be delayed.^{125,126}

With therapeutic dosing, salicylic acid undergoes first-order hepatic metabolism and <10% of salicylate is excreted unchanged by the kidneys. Its elimination half-life is 2 to 4 hours.^{127,128} In acute overdose, protein binding decreases from 90% to 50%, the V_D increases, and metabolism becomes saturated, leading to a prolonged elimination half-life (>30 hours).¹²⁹⁻¹³¹ Salicylic acid is a weak acid (pKa of 3.0). It exists predominantly in the nonionized form in the plasma at physiologic pH:



Nonionized particles cross cell membranes including the blood-brain barrier more readily than the ionized form. Acidemia drives the above reaction to the right, increasing CNS toxicity.¹³² Alkalinization drives the reaction to the left, reducing cell toxicity and favoring urinary elimination.

CLINICAL PRESENTATION AND DIAGNOSTIC TESTING

Salicylate poisoning can be acute or chronic. Acute ingestions >150 mg/kg lead to mild-to-moderate toxicity while >500 mg/kg is potentially lethal. The therapeutic range is 10 to 30 mg/dL (0.7–2.2 mmol/L). Toxic symptoms are usually seen at concentrations exceeding 75 mg/dL (5.4 mmol/L) and high risk of lethality >10 mg/dL (7.2 mmol/L).

Acute salicylate ingestion causes nausea, vomiting, and decreased gastric motility. Acid-base abnormalities occur, with the classical finding being mixed respiratory alkalosis and high anion gap metabolic acidosis. Salicylate stimulates the respiratory center in the brainstem, leading to a fall in PCO_2 .¹³³⁻¹³⁵ Metabolic acidosis is caused by accumulation of organic acids and lactic acid production from increased minute ventilation and uncoupling of mitochondrial oxidative phosphorylation.^{133,136}

Salicylates cause a variety of neurologic symptoms including tinnitus, central hyperthermia, vertigo, agitation, delirium, hallucination, or coma.^{137,138} As stupor progresses, there may be blunting of the respiratory response, decreasing blood pH, and increasing salicylate entry to the CNS.^{132,139} Tinnitus occurs at salicylate concentration of 30 mg/dL

(2.2 mmol/L). Early salicylate poisoning may present with hyperglycemia due to glycogenolysis, gluconeogenesis, and decreased peripheral use. Hypoglycemia occurs later and is a marker of severe poisoning.¹⁴⁰

Chronic poisoning from regular therapeutic salicylate doses can occur, especially when kidney function is decreased. Symptoms such as confusion and pulmonary edema are more prominent than noted at the same concentration following acute ingestion.¹⁴¹⁻¹⁴⁴

The diagnosis of salicylate poisoning is suspected from the history, clinical findings, and metabolic abnormalities. An elevated anion gap with respiratory alkalosis should prompt salicylate measurement. Because absorption may be erratic or prolonged, serial measurements (every 2–4 hours) are required to identify the peak concentration. While salicylate concentrations reflect the risk of poisoning, treatment is also prompted by symptoms. The Done nomogram is no longer used because of its poor predictive value.¹⁴⁵

TREATMENT

General principles of poisoning management apply to salicylates. However, patients are dependent on high-minute ventilation and high serum pH to reduce salicylate entry into the CNS. Endotracheal intubation should be performed only if necessary and by an experienced clinician.¹⁴⁶ Ventilator settings should replicate the patient's respiratory pattern before intubation, although difficult because of auto-PEEP.

Further therapy aims at decreasing absorption and increasing elimination of salicylates. MDAC decreases absorption and increases elimination^{147,148} but is less efficacious for enhanced elimination than urine alkalinization.^{5,6,149,150} Serum and urine alkalinization are crucial. For salicylate poisoning, urine alkalinization reduces the elimination half-life from 19 to 5 hours.¹⁵¹ Because the ionization constant (pKa) is a logarithmic function, small changes in urine pH will have a large effect on salicylate elimination.^{152,153} The bicarbonate infusion is titrated to a urine pH of 7.5 and continued until the serum salicylate concentration is <30 mg/dL (2.2 mmol/L). Alkaline urine cannot be produced in hypokalemia since kidney reabsorption of potassium occurs via the H^+/K^+ exchange pump in exchange for hydrogen. For this reason, and because kaliuresis complicates bicarbonate therapy, potassium should be monitored and hypokalemia aggressively corrected. Preemptive potassium supplementation is recommended if serum potassium is <4 mmol/L prealkalinization. Alkalinization may be contraindicated in AKI or pulmonary edema, when IHD is preferred. Neither MDAC nor urine alkalization replace ECTR in severe poisonings.

Salicylate is highly dialyzable due to its low V_D , low MW, and saturation of protein binding at high plasma concentrations.¹⁵⁴⁻¹⁵⁶ IHD maximizes salicylate removal and corrects acid-base status.¹⁵⁷⁻¹⁵⁹ It is noteworthy that most deaths occur before ECTR initiation.⁸⁰

Indications for ECTR include any of the following attributable to salicylate poisoning¹²:

- Neurologic symptoms (confusion, seizures, coma)
- New hypoxemia requiring supplemental oxygen (e.g., due to pulmonary edema)
- pH <7.25
- Serum salicylate concentration >90 mg/dL (6.5 mmol/L)

- Impaired kidney function with salicylate concentration >90 mg/dL (6.5 mmol/L)
- Clinical deterioration despite appropriate treatment

Treatments should be maintained until salicylate concentrations are below 19 mg/dL (1.4 mmol/L) with clinical improvement.^{12,160}

LITHIUM

Lithium has a narrow therapeutic range (0.4–0.8 mmol/L) and major side effects are rapidly seen once supratherapeutic concentrations are reached.^{161–165}

TOXICOLOGY AND TOXICOKINETICS

The absorption of lithium is rapid at therapeutic doses. Peak blood concentrations are usually delayed 8 to 12 hours post-ingestion in acute poisoning. Lithium is unbound to proteins and has a V_D of 0.7 to 0.8 L/kg. It distributes slowly into the brain, explaining the absence of initial CNS effects after an acute overdose.¹⁶⁶ Lithium is eliminated by the kidneys, with 80% being reabsorbed in the tubules, so total body clearance is $\approx 20\%$ of GFR. Lithium reabsorption follows that of sodium, so salt-depleted states (e.g., volume depletion) increase lithium retention. The half-life of lithium with therapeutic use is usually 18 to 24 hours but can be prolonged in the elderly, chronic users, and CKD.^{167,168} With a normal kidney function, the elimination half-life is approximately 10 hours in acute overdose (due to distribution into extravascular tissues) versus 30 to 40 hours in chronic overdose.¹⁶⁹

CLINICAL PRESENTATION AND DIAGNOSTIC TESTING

Lithium overdose is defined as “acute” after a single exposure in a naïve patient and as “chronic” after dosing or prescribing errors or when lithium clearance becomes impaired.¹⁷⁰ Patients with acute poisoning have few symptoms at concentrations of 4.0 mmol/L, while clinical toxicity can occur in chronic poisonings with near therapeutic concentrations.^{168,171} In acute poisoning, GI symptoms (nausea, vomiting, or diarrhea) and nonspecific cardiac conduction delay are predominant, although life-threatening dysrhythmias are uncommon. In chronic poisoning, symptoms are mostly neurologic, ranging from coarse tremor or dysarthria to lethargy, seizures, hyperthermia, coma, and death.^{172,173} A protracted neurologic course can be seen after severe poisoning.¹⁶⁷ Some patients develop a syndrome of irreversible lithium-effectuated neurotoxicity (SILENT), which relates to cerebellar and cognitive deficits.¹⁷⁴

TREATMENT

Therapy is guided by the type of poisoning (acute vs. chronic), lithium concentrations, symptoms, and kidney function.^{169,171}

Hyperthermia, dysrhythmias, and seizures should be treated according to standard protocols. Volume contraction favors proximal lithium reabsorption and should be promptly corrected with isotonic saline. Following resuscitation, hypotonic solutions or free water may be required for significant hyponatremia resulting from lithium-induced nephrogenic diabetes insipidus.

Whole bowel irrigation may prevent absorption of sustained-release formulations.^{175,176} Oral sodium polystyrene sulfonate (Kayexalate), a cation-exchanger used for hyperkalemia, can bind unabsorbed lithium and enhance its elimination^{29,177,178} and can be considered in patients without contraindications with mild-to-moderate symptoms for whom ECTR is delayed or not offered.^{29,179}

Lithium is readily dialyzable and IHD is the modality of choice, with clearances of 180 mL/min with modern filters.^{180–183} Lithium concentrations often rebound after IHD,^{167,184} but this should not be concerning, as lithium CNS concentrations continue to decrease during this time,⁸⁴ unless there is ongoing absorption from the gut. CRRT provides clearances of approximately 25% of that by IHD.^{181,185–189}

Indications for ECTR include any of the following¹⁶:

- Severe neurologic features (central hyperthermia, seizures, and/or altered consciousness)
- Serum Li concentration >5 mmol/L
- Kidney impairment with symptoms and serum Li concentration >4 mmol/L
- Life-threatening dysrhythmias

The threshold for ECTR is lower with impaired kidney function or if volume repletion is not tolerated.^{167,190}

VALPROIC ACID

Valproic acid is used to treat seizures, migraine, and mood disorders. Although acute valproic acid poisoning often results in mild self-limiting CNS depression, toxic effects and death are reported.^{191,192}

TOXICOLOGY AND TOXICOKINETICS

Valproic acid has a small MW (144.21 Da) and V_D (0.2 L/kg) and a high bioavailability. Serum concentrations peak 1 to 13 hours after ingestion, depending on the preparation and dose.¹⁹³ Therapeutic serum concentrations range from 50 to 100 mg/L (347–693 $\mu\text{mol/L}$). Protein binding of valproic acid is saturable in overdose, being 90% at therapeutic concentrations and 35% at 300 mg/L (2079 $\mu\text{mol/L}$).¹⁹⁴

The liver rapidly metabolizes valproic acid. It undergoes glucuronic acid conjugation (70%) and β - and ω -oxidation to various metabolites, while <3% is excreted unchanged in urine.¹⁹⁴ In overdose, more is metabolized by CYP450-mediated ω -oxidation to 5-OH-VPA and 4-en-VPA, which are responsible for some of the toxic effects.

CLINICAL PRESENTATION AND DIAGNOSTIC TESTING

Most valproic acid overdoses are well tolerated. Toxicity is more likely following ingestions >200 mg/kg.^{192,195} Acute poisoning is manifested by GI distress (nausea, vomiting, diarrhea); CNS abnormalities (confusion, obtundation, coma with respiratory failure); hypotension; and elevated transaminase concentrations. Free and total valproic acid serum concentrations poorly correlate with the severity of poisoning, but most patients with total concentrations greater than 180 mg/L (1247 $\mu\text{mol/L}$) develop some degree of CNS depression.¹⁹⁶ Hyperammonemia can occur at any concentration,¹⁹⁷ and when markedly elevated, it can contribute to encephalopathy, cerebral edema, and death.

At very high serum concentrations (>1000 mg/L or 6930 $\mu\text{mol/L}$), complications include high anion gap metabolic acidosis, hyponatremia, hypotension, hypocalcemia, pancreatitis, noncardiogenic pulmonary edema, bone marrow suppression, and AKI.^{198,199} Diagnosis is based on a history of exposure, symptoms, and serum valproate concentration.

TREATMENT

Treatment begins with stabilization of respiratory and cardiovascular function. Activated charcoal should be administered, particularly within 1 hour of exposure, but can be useful beyond that period due to the prolonged absorption phase in overdose. Supportive data are limited regarding L-carnitine as a treatment for hyperammonemic encephalopathy.^{200,201} If used, intravenous L-carnitine can be administered with a loading dose of 100 mg/kg, followed by infusions of 50 mg/kg every 8 hours, up to a maximum of 3 g per dose until ammonia levels are decreasing.

IHD clears ammonia, corrects metabolic acidosis, and removes valproic acid in overdose due to saturation of plasma protein binding.^{193,202}

Indications for ECTR include any of the following attributable to valproate toxicity¹³:

- Serum valproate concentration >900 mg/L (6250 $\mu\text{mol/L}$)
- Shock
- Cerebral edema, coma, or respiratory depression requiring mechanical ventilation
- Acute hyperammonemia
- $\text{pH} \leq 7.10$

Rebound of valproic acid concentrations occurs 5 to 13 hours after IHD, requiring additional sessions.¹⁹⁴ Hemoperfusion has been successfully used in cases but is limited by early column saturation.¹⁹¹ “In-series” IHD-hemoperfusion offers marginal advantage over IHD in terms of efficacy but is offset by the added cost and potential complications.²⁰³ CRRT is less effective and should only be used if IHD is unavailable or if cerebral edema is present.^{13,204} Albumin dialysis, TPE, and PD are inferior therapeutic options in valproate poisoning and are not recommended.^{205,206}

CARBAMAZEPINE

Carbamazepine is used for seizures, pain management, and mood disorders.

TOXICOLOGY AND TOXICOKINETICS

Carbamazepine inhibits sodium channels, neuronal depolarization, and glutamate release and appears to have anticholinergic effects at high concentrations.

Carbamazepine is available in immediate and modified-release formulations and is characterized by erratic and incomplete absorption, which is exacerbated in overdose due to pharmacobezoar formation and ileus.²⁰⁷ Carbamazepine has small MW (236 D), small-to-moderate V_D (1.2 L/Kg), and a protein binding of approximately 75% , which does not decrease much in overdose.²⁰⁸ It undergoes hepatic metabolism into many metabolites, the most important being carbamazepine-10,11-epoxide,

which is pharmacologically active. Carbamazepine induces its own metabolism, which increases endogenous clearance with chronic use. The therapeutic concentration range is 4 to 12 mg/L (16.9 – 50.8 $\mu\text{mol/L}$).

CLINICAL PRESENTATION AND DIAGNOSTIC TESTING

Carbamazepine toxicity frequently presents with neurologic, cardiovascular, and anticholinergic symptoms, which may be delayed because of its erratic absorption. Mild toxicity (≈ 30 mg/L or 127 $\mu\text{mol/L}$) presents as drowsiness, nystagmus, tachycardia, hyperreflexia, or dysmetria. In severe exposures (>40 mg/L or 169 $\mu\text{mol/L}$), lethargy, seizure, coma, QRS prolongation, hypotension, and ileus may develop; death is unusual. Agranulocytosis and SIADH are typically not seen in acute poisonings.

Diagnosis relies on the history, clinical findings, and laboratory testing. Serial measurements of serum carbamazepine concentrations are required because the time to peak may be beyond 24 hours. Concentrations should be obtained every 4 to 6 hours until a definite downward trend is seen.

TREATMENT

Most patients can be managed with supportive care including ventilatory support, benzodiazepines for seizures, vasopressors for hypotension, and sodium bicarbonate for sodium channel blockade. GI decontamination is administered even if the patient presents late after ingestion due to the prolonged absorption phase but is contraindicated if ileus is present. MDAC can enhance carbamazepine clearance and may reduce the duration of coma and need for mechanical ventilation.^{5,209-212}

Severe carbamazepine poisonings can be treated using ECTR, providing better and more predictable clearance than MDAC.²¹³⁻²¹⁵ Carbamazepine clearance with IHD compares with that of hemoperfusion and is therefore the ECTR of choice, given all the known advantages stated earlier.^{10,216,217} There are limited data on the ECTR clearance of the toxic metabolite carbamazepine-10,11-epoxide.^{208,218-220} CRRT, TPE, and albumin dialysis have been used but provide lesser clearances.^{221,222}

Indications for ECTR include any of the following attributable to carbamazepine toxicity¹⁰:

- Prolonged coma
- Seizures, cardiovascular instability, or other signs unresponsive to supportive care
- Rising carbamazepine concentrations despite MDAC

BARBITURATES

Barbiturates are CNS depressants used as sedatives, hypnotics, anxiolytics, and anticonvulsants. Phenobarbital is the most commonly used barbiturate worldwide.²²³

TOXICOLOGY AND TOXICOKINETICS

Absorption kinetics are variable and influenced by the dose, presence of ileus, and coingested drugs. Barbiturates are small molecules with low protein binding. Their metabolism is mostly hepatic, but phenobarbital undergoes some renal excretion. Chronic barbiturate use induces its own metabolism, increasing endogenous clearance.

An acute oral dose of 1 g of most barbiturates will cause serious toxicity. For phenobarbital, death can result in ingestions >2 g²²⁴ or concentrations >80 mg/L (345 μ mol/L),²²⁵ without supportive care. The following discussion focuses on phenobarbital since it is the most encountered, and recommendations also apply to primidone because it is partially metabolized to phenobarbital.

CLINICAL PRESENTATION AND DIAGNOSTIC TESTING

Phenobarbital has prolonged clinical effects,^{226,227} which progress from sedation in mild exposures to apnea, hypotension, circulatory collapse, hypothermia, and coma in large ingestions.^{223,224} Concomitant AKI, cardiac, or pulmonary disease increases the clinical sensitivity to barbiturates.²²⁸ Early deaths after barbiturate ingestion are caused by respiratory and cardiovascular collapse, whereas later deaths are caused by acute lung injury, ventilator-acquired pneumonia, cerebral edema, or multiorgan failure, particularly when there is delayed presentation for medical care.

TREATMENT

Supportive measures including rewarming, hydration, and vasopressors are usually sufficient. Patients may be profoundly comatose and require mechanical ventilation for many days. MDAC increases the clearance of phenobarbital²²⁹⁻²³¹ and is the first-line treatment for enhanced elimination. Urinary alkalization can increase phenobarbital renal clearance by twofold to threefold.^{232,233} However, because renal clearance of phenobarbital is low (<3 mL/min), alkalization has lesser impact.²³² A controlled trial indicated that MDAC was superior to MDAC plus urinary alkalization, which was superior to urinary alkalization.²³¹

No randomized studies have evaluated the effect of ECTR in humans, although the mortality rate was lower in an uncontrolled group with hemoperfusion.²³⁴ Modern high-flux dialyzers provide clearances at least equal to hemoperfusion cartridges.

Indications for ECTR include any of the following (in particular for phenobarbital poisoning)^{8,223}:

- Coma with respiratory depression
- Hypotension not responding to vasopressors
- Inefficiency of MDAC at reducing serum concentrations

When treating barbiturate poisoning in chronic users, there is a risk of precipitating barbiturate withdrawal when concentrations fall below the therapeutic range. This can be minimized by stopping ECTR and other treatments when the phenobarbital concentration is therapeutic, after which it may be prudent to reinstitute barbiturate treatment.

PHENYTOIN

Phenytoin is a first-line treatment of epilepsy. Phenytoin toxicity can occur readily due to its narrow therapeutic range, although phenytoin-related deaths are rare.

TOXICOLOGY AND TOXICOKINETICS

Phenytoin has high protein binding (90 to 95%), which decreases to 70% in the presence of CKD or hypoalbuminemia but remains almost unchanged in overdose.

Phenytoin has erratic oral bioavailability, especially in overdose.^{235,236} Phenytoin is mainly metabolized by the liver.

The elimination half-life with oral therapeutic dosing is 14 to 22 hours but significantly longer in overdose. Therapeutic serum concentrations are 10 to 20 mg/L (40 – 80 μ mol/L) and the potentially lethal dose in adults is 2 to 5 g.

CLINICAL PRESENTATION AND DIAGNOSTIC TESTING

Phenytoin toxicity is predominantly neurologic, with severity loosely correlating with serum concentrations. Between 20 and 40 mg/L (80 – 160 μ mol/L), nystagmus, ataxia, and mild CNS depression are apparent,²³⁷ which progress to lethargy, confusion, hypotension, coma, and seizures >40 mg/L (160 μ mol/L). Cardiovascular toxicity is unusual with oral overdoses, but short-lived atrioventricular delays and bradycardia may occur following rapid intravenous infusion.²³⁷ Death caused by respiratory or circulatory depression is rare.

Total serum phenytoin concentrations should be obtained in all suspected overdoses. Some conditions (uremia, extremes of age, or concomitant use of agents) displace phenytoin from its albumin-binding site. The free phenytoin concentration should be measured if at all possible in such cases, and toxicity is apparent with free (unbound) phenytoin concentrations >2.1 mg/L (8.3 μ mol/L).²³⁸

TREATMENT

Supportive treatment is usually sufficient since most patients have an excellent outcome, although some may have a prolonged length of stay. Benzodiazepines are preferred for seizures. Due to prolonged absorption in overdose, gastrointestinal decontamination should also be considered in delayed presentations. Multiple-dose activated charcoal significantly reduces the elimination half-life of phenytoin.^{28,239,240}

Experience with ECTRs is limited. Given the high protein binding of phenytoin, HP or TPE has been historically favored and used successfully.²⁴¹⁻²⁴⁴ Surprisingly, IHD can enhance the elimination of phenytoin, perhaps due to its low binding constant to albumin, which ensures a constant pool of freely diffusible unbound phenytoin.²⁴⁵⁻²⁴⁷ This finding is attributable to high-flux, high-efficiency dialyzers—older ECTR apparatus did not significantly increase phenytoin clearance.²⁴⁸

Neither PD nor CRRT has a role in phenytoin poisoning,²⁴⁹⁻²⁵² and data are uncertain for albumin dialysis.⁶⁷ In the case of actual or anticipated toxicity, with persistently elevated phenytoin concentration, IHD or HP can be considered.¹⁷

METFORMIN

Metformin is the most popular antidiabetic drug worldwide.

TOXICOLOGY AND TOXICOKINETICS

Metformin has a large V_D (3 L/Kg) and high endogenous clearance from renal tubular secretion, with clearance up to 500 mL/min in patients with normal kidney function. The toxic dose in acute ingestions is likely to exceed 100 mg/kg.

CLINICAL PRESENTATION AND DIAGNOSTIC TESTING

Metformin toxicity manifests as GI symptoms (e.g., abdominal pain, diarrhea, nausea, and vomiting), metabolic acidosis

with hyperlactatemia, and hypotension. Hypoglycemia, hypothermia, altered mental status, and acute pancreatitis have also been reported.^{253,254} Metformin-associated lactic acidosis (MALA) usually occurs in the presence of underlying conditions, particularly AKI or CKD. MALA is variably defined as an arterial pH ≤ 7.35 and an elevated lactate concentration (e.g., >45 mg/dL or 5 mmol/L) from acute or chronic metformin poisoning.²⁵⁵⁻²⁵⁷ The extent to which metformin causes hyperlactatemia is controversial; some sources claim that the association is coincidental,^{258,259} while others suggest that metformin is causative.²⁶⁰ The finding of asymptomatic patients who develop toxicity shortly after acute toxic ingestion gives credence to the latter hypothesis.²⁶⁰⁻²⁶³ Some, but not all, studies have reported a correlation between metformin concentrations and hyperlactatemia^{258,260,264-266}

TREATMENT

Management is supportive and includes intravenous hydration, treating concomitant and exacerbating conditions, normalizing acid-base abnormalities, and reducing metformin concentrations. Activated charcoal should be considered within 1 to 2 hours of an acute large ingestion. Patients who develop metabolic acidosis with hyperlactatemia, hypoglycemia, or other signs of metformin toxicity should be admitted to the ICU.

Severely acidotic patients (pH ≤ 7.1) should receive intravenous sodium bicarbonate, although treatment can be limited by hypernatremia and volume overload. ECTRs correct acidosis faster than intravenous bicarbonate, clear lactate and metformin, and treat volume overload and uremia. A dramatic improvement from ECTR has been described in some acutely poisoned patients soon after IHD initiation, suggesting that the benefit might be attributable to pH correction rather than metformin removal.²⁶⁷ Studies note similar mortality rates between patients who did and did not receive ECTR, but the IHD group was sicker at baseline, which may support a clinical benefit of ECTR.^{255,257,268,269} Metformin distributes in a smaller V_D in AKI and is more readily removed by ECTR in these cases.²⁷⁰⁻²⁷²

Indications for ECTR include any of the following attributable to metformin toxicity⁷¹:

- Lactate concentration >180 mg/dL (20 mmol/L)
- Arterial pH ≤ 7.1
- Failure of supportive therapy in severe MALA
- Shock, impaired kidney function, or coma

Recent research has confirmed the poor outcomes related to these conditions.²⁷³ IHD is preferred over CRRT due to higher metformin and lactate clearances.^{71,78,274} More lactate is removed following 6 hours of IHD than 24 hours of continuous venovenous hemodiafiltration,²⁷⁵ although both remain inferior to normal physiological processes.²⁷⁶ Although IHD provides superior metformin and lactate clearance, CRRT may be used in some settings.^{257,277} Hemoperfusion and TPE are not recommended.²⁷⁸

Endpoints for ECTR cessation include normalization of blood pH and lactate. Shortened ECTR sessions can result in a marked rebound in lactate concentration,²⁷⁹ which may be reduced by sequentially using CRRT after IHD.²⁷⁸

PARAQUAT

Paraquat (1,1'-dimethyl-4,4'-bipyridylium dichloride) is a fast-acting herbicide. Oral ingestion of paraquat is extremely toxic, with high morbidity and mortality (50%–90%) from exposures as low as 20 mL. Its access is restricted in some parts of the world but not in many developing countries.²⁸⁰

TOXICOLOGY AND TOXICOKINETICS

Plasma paraquat concentrations peak by 2 hours post-ingestion, with maximal tissue distribution over the next 6 hours.^{281,282} Paraquat is distributed to most organs, especially the lungs, kidneys, and liver. It is predominantly excreted by the kidneys, with an elimination half-life of 12 hours with normal kidney function, and >48 hours with impaired kidney function.^{282,283}

Paraquat catalyzes the formation of reactive oxygen species (ROS) through redox cycling. The resultant oxidative stress causes extensive cell injury and necrosis and a secondary inflammatory reaction.^{281,284,285}

CLINICAL PRESENTATION AND DIAGNOSTIC TESTING

The clinical course depends on the dose and timing of presentation to a hospital. Ingestions between 20 and 40 mg/kg will usually cause AKI and multiorgan failure with death one to several weeks post-exposure, and ingestions >40 mg/kg cause death within 3 days.²⁸⁶⁻²⁸⁹ Most deaths are caused by acute respiratory distress syndrome (ARDS)-like presentation, progressing to pulmonary fibrosis^{281,290} or corrosion to mucosal membranes and esophageal perforation, hepatic injury, and shock. Paraquat does not typically cause CNS effects.

Serum paraquat concentration and validated nomograms predict mortality, although assays are rarely available.²⁹¹⁻²⁹⁶ A rapid and inexpensive qualitative test can be obtained by adding sodium dithionite to urine. If the urine changes from yellow to blue, it confirms recent paraquat exposure (the more intense the blue the higher the exposure).

TREATMENT

Any suspected exposure warrants prompt assessment and aggressive management. After stabilization, volume resuscitation should be provided. Oxygen should be used only if required for hypoxemic patients because it may promote cellular toxicity from paraquat. Paraquat is absorbed quickly, limiting the efficacy of decontamination; activated charcoal should be administered within 1 hour of ingestion. No specific antidote is available.

The poor prognosis related to paraquat poisoning has prompted the use of experimental immunosuppressive or antiinflammatory treatments, which have not altered clinical outcomes. These include cyclophosphamide and corticosteroids (dexamethasone).²⁹⁷

Because paraquat is a small molecule, unbound to plasma proteins and with a small V_D , it is removable by ECTRs (clearance >120 mL/min). However, paraquat distributes quickly to tissues, especially the lungs. Once the toxic process of free-radical generation is established, ECTR is unlikely to mitigate disease progression. This concept has been confirmed in animal studies: half of the dogs exposed to a lethal dose survived with hemoperfusion when initiated 2 hours after exposure, whereas all died when initiated at

12 hours.²⁹⁸ IHD or hemoperfusion is mostly reserved for early presentation, especially within 4 hours.^{283,299,300} Other ECTRs are of limited utility.²⁸³

THEOPHYLLINE

Theophylline is a methylxanthine rarely used nowadays to treat asthma, chronic obstructive pulmonary disease, and infant apnea.

TOXICOLOGY AND TOXICOKINETICS

Theophylline is a nonselective inhibitor of phosphodiesterase and a nonselective antagonist of the adenosine receptor. This results in smooth muscle relaxation, catecholamine release, bronchodilation, peripheral vasodilatation, and inotropic and chronotropic activation.³⁰¹

Theophylline is available in immediate and slow-release formulations and is completely absorbed in the GI tract, with a small V_D . Theophylline is mainly metabolized in the liver by CYP1A2. Endogenous metabolism varies depending on comorbidities, smoking status, and other drugs.³⁰² At therapeutic serum concentrations (10–20 mg/L or 55–110 $\mu\text{mol/L}$), theophylline exhibits first-order kinetics; however, at toxic concentrations, the kinetics change to zero order and a small increase in dosage can significantly increase serum concentrations.³⁰³

CLINICAL PRESENTATION AND DIAGNOSTIC TESTING

Theophylline toxicity is seen from acute ingestions (self-poisoning or error) or chronic use (error, or decreased clearance). The distinction is important since symptoms are more severe in chronic poisoning than in acute poisoning for the same serum theophylline concentration.³⁰⁴ Even at therapeutic concentrations, nearly one-third of patients show signs of mild intoxication. Caffeine, an active metabolite of theophylline, shares many of its toxic effects.

At serum concentrations >30 mg/L (167 $\mu\text{mol/L}$), symptoms include vomiting, diarrhea, irritability, tremor, and tachycardia. At serum concentrations >90 mg/L (500 $\mu\text{mol/L}$), symptoms include seizures, hyperthermia, coma, hypotension, ventricular tachycardia, rhabdomyolysis, and AKI. Death most often results from cardiorespiratory collapse or hypoxic encephalopathy following cardiac dysrhythmias or generalized seizures.^{304–306} Hypokalemia, caused by transcellular potassium shifts, can be severe and is most common with acute overdose.^{305,307}

TREATMENT

Supportive measures include volume repletion and vasopressors for hypotension. Tachycardia can be corrected by β -adrenergic antagonists but should be used with caution in patients susceptible to bronchospasm.^{308,309} Seizures can be difficult to control; benzodiazepines are first-line therapy, but propofol, barbiturates, and even neuromuscular paralysis may be required if refractory. Multiple-dose activated charcoal increases theophylline clearance, which is sometimes limited by intractable emesis, which is reduced by intubation.^{310–312}

In one retrospective cohort study, the group treated with hemoperfusion or IHD had significantly shorter clinical toxicity compared with supportive care, despite being sicker at presentation.³¹³ IHD provides similar clearance to

hemoperfusion (>150 mL/min)^{314–316} but corrects electrolyte disorders and is associated with less complications.⁴⁴ Continuous techniques have also been used with acceptable results despite their inferior clearances.^{317,318} PD and TPE are not recommended. ET is an option for neonates.³¹⁹

Indications for ECTR include any of the following attributable to theophylline toxicity¹⁴:

- Clinical deterioration despite optimal therapy
- Serum theophylline concentration >100 mg/L (555 $\mu\text{mol/L}$)
- Chronic poisoning with serum theophylline concentration >60 mg/L (333 $\mu\text{mol/L}$)
- Refractory seizures, shock, life-threatening dysrhythmias, inability to administer charcoal because of intractable vomiting, or extremes of age (<6 months or >60 years)
- A rising serum theophylline concentration despite optimal therapy or if multiple-dose activated charcoal cannot be administered

ECTR should be continued until clinical improvement and serum concentration is <15 $\mu\text{g/mL}$ (83 $\mu\text{mol/L}$). Rebound is usually minor but should be monitored after ECTR in case ongoing absorption occurs.

ACETAMINOPHEN

Acetaminophen (paracetamol), an analgesic and antipyretic available without prescription in various forms and combinations, is a major cause of self-poisoning.

TOXICOLOGY AND TOXICOKINETICS

The actions of acetaminophen remain poorly defined. Analgesic and antipyretic activity occur at serum concentrations between 10 and 20 mg/L (66–132 $\mu\text{mol/L}$).³²⁰ Acetaminophen is available in immediate and extended-release preparations and has high oral bioavailability. Serum concentrations usually peak within 4 hours of ingestion but may be delayed by many hours with massive overdoses or from sustained-release formulations. Protein binding and V_D are small and unchanged in overdose. Acetaminophen undergoes extensive hepatic metabolism, where 90% is conjugated to inactive metabolites, and the remainder is oxidized mostly by CYP2E1 to N-acetyl-p-benzoquinone imine (NAPQI). Glutathione combines with NAPQI to form a nontoxic complex eliminated in the urine. In overdose, there is an accumulation in NAPQI and glutathione depletion, allowing NAPQI to bind to proteins in hepatocytes, inducing injury and liver cell necrosis.³²¹ The lowest toxic dose following acute poisoning is 10 g in single dosing over 8 hours.³²²

CLINICAL PRESENTATION AND DIAGNOSIS TESTING

Patients may experience symptoms including nausea, vomiting, anorexia, malaise, or abdominal pain in the first 24 hours. An increase in transaminases typically occurs 12 to 24 hours after ingestion and peaks 2 to 3 days after ingestion in mild-to-moderate poisoning but later in severe hepatotoxicity. Severe cases can evolve into fulminant liver failure with encephalopathy, coagulopathy, hypoglycemia, and death. AKI can occur³²³ and generally recovers after 1 to 3 months.

The risk of acute acetaminophen poisoning is estimated by plotting a serum acetaminophen concentration

relating to the time of ingestion on the Rumack-Matthew nomogram.³²⁴ A point below the “treatment line” indicates that the patient has a low risk of developing hepatotoxicity.

TREATMENT

Management of acetaminophen toxicity is reviewed elsewhere³²⁵ but focuses on decontamination for early presentations and administration of the antidote, N-acetylcysteine (NAC).³²⁶ NAC is a glutathione donor, increasing the capacity to detoxify NAPQI.³²⁷ It is most efficacious when initiated within 8 hours. NAC is prescribed if there is liver injury following a history of known or possible acetaminophen poisoning, if the concentration is above the treatment line on the nomogram, or even in cases of possible acetaminophen poisoning, due to the low risk of NAC.

Although acetaminophen is amenable to removal by ECTR, it is rarely necessary, given the safety, low cost, efficacy, and availability of NAC. IHD or CRRT can be considered for patients with severe AKI according to usual criteria or following massive ingestions with mitochondrial toxicity (coma, lactic metabolic acidosis, and/or cardiovascular instability).^{11,328,329} These patients can be identified by an early onset of these signs with massive exposure versus hepatotoxicity with the delayed presentation previously described. In these cases, the risk of ECTR removing NAC can be overcome by doubling its infusion rate (some sources suggest an increased dose of NAC with massive acetaminophen poisonings anyway).^{74,330} MARS and SPAD have been used for liver support as bridges to liver transplantation or pending native liver recovery.³³¹⁻³³⁵ However, their limited availability, bleeding complications, and high-cost warrant further investigation.

METHOTREXATE

Methotrexate is used for a variety of indications including cancers, rheumatoid arthritis, and inflammatory bowel disease. Methotrexate inhibits dihydrofolate reductase, an enzyme necessary for cell division.

TOXICOLOGY AND TOXICOKINETICS

Oral absorption of methotrexate is limited by a saturable intestinal absorption mechanism for doses over 30 mg/m². For this reason, single acute oral ingestions of methotrexate do not cause toxicity in patients with normal renal function.³³⁶

Methotrexate toxicity is most often related to dosing errors in oral dosing or in susceptible patients with parenteral administration. Methotrexate (molecular weight of 454 Da) has a small volume of distribution (0.4–1.0 L/kg) and is 50% bound to plasma proteins, regardless of serum concentration. Methotrexate is primarily excreted unchanged (80%–90%) in the urine by passive glomerular filtration and active tubular secretion, the remainder either excreted in bile or metabolized in the liver.

CLINICAL PRESENTATION AND DIAGNOSTIC TESTING

Methotrexate and its metabolites can precipitate in renal tubules, causing crystal nephropathy and AKI, especially when the concentration exceeds 10 µmol/L, 24 hours after

parenteral administration. Methotrexate can also cause bone marrow suppression, mucositis and stomatitis, liver damage, and neurotoxicity.

TREATMENT

Aggressive hydration is indicated for patients presenting with methotrexate toxicity. Because methotrexate is a weak acid (pK_a = 5), and because it is eliminated largely unchanged in the urine, urinary alkalization with intravenous sodium bicarbonate (target urine pH ≈ 8) will enhance methotrexate elimination and may help to prevent AKI.^{6,337}

Leucovorin (folinic acid; an active form of folate) can limit the bone marrow and gastrointestinal toxicity of methotrexate by bypassing the effect of methotrexate on dihydrofolate reductase. Leucovorin “rescue” is most beneficial when administered promptly after methotrexate exposure and should always be given to patients after a supratherapeutic exposure. Other supportive measures include transfusion of blood components for established cytopenias, antiemetics, and nutritional support for stomatitis. Dose recommendations and precise application are covered elsewhere.³³⁸

Glucarpidase (carboxypeptidase G₂; CPDG₂) is a recombinant bacterial enzyme that quickly catabolizes methotrexate to inactive metabolites. This antidote is used in patients with AKI and a serum methotrexate concentration of >50 µmol/L at 24 hours or >30 µmol/L at 36 hours but has no role after oral toxicity. Glucarpidase is expensive (>US\$50,000) and either unavailable or restricted in most countries. There is controversy regarding its efficacy, in particular because its activity is limited to the plasma, and it also catabolizes leucovorin.

Although ECTRs can enhance elimination of methotrexate, they are rarely, if ever, indicated in methotrexate toxicity.²⁰ Firstly, ECTRs only remove methotrexate in plasma (much less efficiently than glucarpidase) and do not influence intracellular stores, which are responsible for toxicity. Further, clinical outcomes do not appear improved by the use of ECTRs. Finally, ECTRs remove lifesaving leucovorin.

OTHERS

There are other conditions and poisoning situations in which enhanced elimination using ECTR may reduce the duration of toxicity. Successful extracorporeal removal is reported with aluminum,³³⁹ atenolol,²² baclofen,²⁶ barium,³⁴⁰ bromate,³⁴¹ carambola,³⁴² cefepime,³⁴³ dabigatran,^{82,344,345} dapsone,³⁴⁶ diethylene glycol,³⁴⁷ fluoride,³⁴⁸ gabapentin,²¹ gentamicin,³⁴⁹ isoniazid,²³ pregabalin,²¹ propylene glycol,³⁵⁰ sotalol,²² thallium,⁷ and vancomycin.³⁵¹

CONCLUSION

Supportive care is sufficient to manage most poisonings. In some cases, ECTR, usually IHD, can reduce the exposure of a patient to the toxic effects of a poison. An understanding of poison toxicokinetics can help a clinician discern what are the timely conditions and circumstances when ECTRs are most likely to be beneficial. EXTRIP have published clinical guidelines to guide the treatment of a number of poisons.

 Visit Elsevier eBooks+ (eBooks.Health.Elsevier.com) for a complete set of references and Board Review Questions.

KEY REFERENCES

- Gummin DD, Mowry JB, Beuhler MC, et al. 2021 annual report of the National Poison Data System((c)) (NPDS) from America's poison centers: 39th annual report. *Clin Toxicol*. 2022;60(12):1381–1643.
- Hoegberg LCG, Shepherd G, Wood DM, et al. Systematic review on the use of activated charcoal for gastrointestinal decontamination following acute oral overdose. *Clin Toxicol*. 2021;59(12):1196–1227.
- Proudfoot AT, Krenzelok EP, Vale JA. Position paper on urine alkalization. *J Toxicol Clin Toxicol*. 2004;42(1):1–26.
- Ghannoum M, Nolin TD, Goldfarb DS, et al. Extracorporeal treatment for thallium poisoning: recommendations from the EXTRIP Workgroup. *Clin J Am Soc Nephrol*. 2012;7(10):1682–1690.
- Mactier R, Laliberte M, Mardini J, et al. Extracorporeal treatment for barbiturate poisoning: recommendations from the EXTRIP Workgroup. *Am J Kidney Dis*. 2014;64(3):347–358.
- Yates C, Galvao T, Sowinski KM, et al. Extracorporeal treatment for tricyclic antidepressant poisoning: recommendations from the EXTRIP Workgroup. *Semin Dial*. 2014;27(4):381–389.
- Ghannoum M, Yates C, Galvao TF, et al. Extracorporeal treatment for carbamazepine poisoning: systematic review and recommendations from the EXTRIP workgroup. *Clin Toxicol*. 2014;52(10):993–1004.
- Gosselin S, Juurlink DN, Kielstein JT, et al. Extracorporeal treatment for acetaminophen poisoning: recommendations from the EXTRIP workgroup. *Clin Toxicol*. 2014;52(8):856–867.
- Juurlink DN, Gosselin S, Kielstein JT, et al. Extracorporeal treatment for salicylate poisoning: systematic review and recommendations from the EXTRIP workgroup. *Ann Emerg Med*. 2015;66(2):165–181.
- Ghannoum M, Laliberte M, Nolin TD, et al. Extracorporeal treatment for valproic acid poisoning: systematic review and recommendations from the EXTRIP workgroup. *Clin Toxicol*. 2015;53(5):454–465.
- Ghannoum M, Wiegand TJ, Liu KD, et al. Extracorporeal treatment for theophylline poisoning: systematic review and recommendations from the EXTRIP workgroup. *Clin Toxicol*. 2015;53(4):215–229.
- Roberts DM, Yates C, Megarbane B, et al. Recommendations for the role of extracorporeal treatments in the management of acute methanol poisoning: a systematic review and consensus statement. *Crit Care Med*. 2015;43(2):461–472.
- Decker BS, Goldfarb DS, Dargan PI, et al. Extracorporeal treatment for lithium poisoning: systematic review and recommendations from the EXTRIP workgroup. *Clin J Am Soc Nephrol*. 2015;10(5):875–887.
- Anseeuw K, Mowry JB, Burdmann EA, et al. Extracorporeal treatment in phenytoin poisoning: systematic review and recommendations from the EXTRIP (extracorporeal treatments in poisoning) workgroup. *Am J Kidney Dis*. 2016;67(2):187–197.
- Mowry JB, Burdmann EA, Anseeuw K, et al. Extracorporeal treatment for digoxin poisoning: systematic review and recommendations from the EXTRIP Workgroup. *Clin Toxicol*. 2016;54(2):103–114.
- Ghannoum M, Gosselin S, Hoffman RS, et al. Extracorporeal treatment for ethylene glycol poisoning: systematic review and recommendations from the EXTRIP workgroup. *Crit Care*. 2023;27(1):56.
- Ghannoum M, Roberts DM, Goldfarb DS, et al. Extracorporeal treatment for methotrexate poisoning: systematic review and recommendations from the EXTRIP workgroup. *Clin J Am Soc Nephrol*. 2022;17(4):602–622.
- Bouchard J, Yates C, Caleslo DP, et al. Extracorporeal treatment for gabapentin and pregabalin poisoning: systematic review and recommendations from the EXTRIP workgroup. *Am J Kidney Dis*. 2022;79(1):88–104.
- Bouchard J, Shepherd G, Hoffman RS, et al. Extracorporeal treatment for poisoning to beta-adrenergic antagonists: systematic review and recommendations from the EXTRIP workgroup. *Crit Care*. 2021;25(1):201.
- Mowry JB, Shepherd G, Hoffman RS, et al. Extracorporeal treatments for isoniazid poisoning: systematic review and recommendations from the EXTRIP workgroup. *Pharmacotherapy*. 2021;41(5):463–478.
- Wong A, Hoffman RS, Walsh SJ, et al. Extracorporeal treatment for calcium channel blocker poisoning: systematic review and recommendations from the EXTRIP workgroup. *Clin Toxicol*. 2021;59(5):361–375.
- Berling I, King JD, Shepherd G, et al. Extracorporeal treatment for chloroquine, hydroxychloroquine, and quinine poisoning: systematic review and recommendations from the EXTRIP workgroup. *J Am Soc Nephrol*. 2020;31(10):2475–2489.
- Ghannoum M, Berling I, Laverne V, et al. Recommendations from the EXTRIP workgroup on extracorporeal treatment for baclofen poisoning. *Kidney Int*. 2021;100(4):720–736.
- Schreiner GE. The role of hemodialysis (artificial kidney) in acute poisoning. *AMA Arch Intern Med*. 1958;102(6):896–913.
- Laverne V, Ouellet G, Bouchard J, et al. Guidelines for reporting case studies on extracorporeal treatments in poisonings: methodology. *Semin Dial*. 2014;27(4):407–414.
- Ouellet G, Bouchard J, Ghannoum M, Decker BS. Available extracorporeal treatments for poisoning: overview and limitations. *Semin Dial*. 2014;27(4):342–349.
- Bouchard J, Roberts DM, Roy L, et al. Principles and operational parameters to optimize poison removal with extracorporeal treatments. *Semin Dial*. 2014;27(4):371–380.
- Ghannoum M, Roberts DM, Hoffman RS, et al. A stepwise approach for the management of poisoning with extracorporeal treatments. *Semin Dial*. 2014;27(4):362–370.
- Bouchard J, Laverne V, Roberts DM, Cormier M, Morissette G, Ghannoum M. Availability and cost of extracorporeal treatments for poisonings and other emergency indications: a worldwide survey. *Nephrol Dial Transplant*. 2017;32(4):699–706.
- Ghannoum M, Bouchard J, Nolin TD, Ouellet G, Roberts DM. Hemoperfusion for the treatment of poisoning: technology, determinants of poison clearance, and application in clinical practice. *Semin Dial*. 2014;27(4):350–361.
- Shannon MW. Comparative efficacy of hemodialysis and hemoperfusion in severe theophylline intoxication. *Acad Emerg Med*. 1997;4(7):674–678.
- Ghannoum M, Nolin TD, Laverne V, Hoffman RS. Blood purification in toxicology: nephrology's ugly duckling. *Adv Chronic Kidney Dis*. 2011;18(3):160–166.
- Laverne V, Nolin TD, Hoffman RS, et al. The EXTRIP (extracorporeal treatments in poisoning) workgroup: guideline methodology. *Clin Toxicol*. 2012;50:403–413.
- Caleslo DP, Liu KD, Wiegand TJ, et al. Extracorporeal treatment for metformin poisoning: systematic review and recommendations from the extracorporeal treatments in poisoning workgroup. *Crit Care Med*. 2015;43(8):1716–1730.
- Laverne V, Hoffman RS, Mowry JB, et al. Why are we still dialyzing overdoses to tricyclic antidepressants? A subanalysis of the npds database. *Semin Dial*. 2016;29(5):403–409.
- Barceloux DG, Krenzelok EP, Olson K, Watson W. American academy of clinical toxicology practice guidelines on the treatment of ethylene glycol poisoning. Ad hoc committee. *J Toxicol Clin Toxicol*. 1999;37(5):537–560.
- Roberts DM, Hoffman RS, Brent J, et al. The serum glycolate concentration: its prognostic value and its correlation to surrogate markers in ethylene glycol exposures. *Clin Toxicol*. 2022;60(7):798–807.
- Barceloux DG, Bond GR, Krenzelok EP, Cooper H, Vale JA. American academy of clinical toxicology practice guidelines on the treatment of methanol poisoning. *J Toxicol Clin Toxicol*. 2002;40(4):415–446.
- Beaulieu J, Roberts DM, Gosselin S, et al. Treating ethylene glycol poisoning with alcohol dehydrogenase inhibition, but without extracorporeal treatments: a systematic review. *Clin Toxicol*. 2022;60(7):784–797.
- Isbister GK, Balit CR, Whyte IM, Dawson A. Valproate overdose: a comparative cohort study of self poisonings. *Br J Clin Pharmacol*. 2003;55(4):398–404.
- Nguyen P, Chevillard L, Gouda AS, et al. L-carnitine does not improve valproic acid poisoning management: a cohort study with toxicokinetics and concentration/effect relationships. *Ann Intensive Care*. 2022;12(1):7.

255. Peters N, Jay N, Barraud D, et al. Metformin-associated lactic acidosis in an intensive care unit. *Crit Care*. 2008;12(6).
273. Trebach J, Mohan S, Gnirke M, Su MK, Gosselin S, Hoffman RS. Retrospective evaluation of management guidelines for extracorporeal treatment of metformin poisoning. *Clin Toxicol*. 2023;61(4):223–227.
280. Gunnell D, Eddleston M, Phillips MR, Konradsen F. The global distribution of fatal pesticide self-poisoning: systematic review. *BMC Publ Health*. 2007;7:357.
297. Gunawardena G, Roberts DM, Buckley NA. Randomized control trial of immunosuppression in paraquat poisoning. *Crit Care Med*. 2007;35(1):330–331; author reply 1.
306. Shannon M. Life-threatening events after theophylline overdose: a 10-year prospective analysis. *Arch Intern Med*. 1999;159(9):989–994.
324. Rumack BH, Peterson RC, Koch GG, Amara IA. Acetaminophen overdose. 662 cases with evaluation of oral acetylcysteine treatment. *Arch Intern Med*. 1981;141(3 Spec No):380–385.
330. Chiew AL, Fountain JS, Graudins A, Isbister GK, Reith D, Buckley NA. Summary statement: new guidelines for the management of paracetamol poisoning in Australia and New Zealand. *Med J Aust*. 2015;203(5):215–218.
336. Chan BS, Dawson AH, Buckley NA. What can clinicians learn from therapeutic studies about the treatment of acute oral methotrexate poisoning? *Clin Toxicol*. 2017;55(2):88–96.
345. Ghannoum M, Nolin TD. What is the role of renal replacement therapy in the setting of dabigatran toxicity? *Semin Dial*. 2014;27(3):223–226.

REFERENCES

- Gummin DD, Mowry JB, Beuhler MC, Spyker DA, Rivers LJ, Feldman R, et al. 2021 annual report of the National Poison Data System ((c)) (NPDS) from America's poison centers: 39th annual report. *Clin Toxicol*. 2022;60(12):1381–1643.
- Daly FFS, Little M, Murray L. A risk assessment based approach to the management of acute poisoning. *Emerg Med J*. 2006;23(5):396–399.
- Hoegberg LCG, Shepherd G, Wood DM, Johnson J, Hoffman RS, Caravati EM, et al. Systematic review on the use of activated charcoal for gastrointestinal decontamination following acute oral overdose. *Clin Toxicol (Phila)*. 2021;59(12):1196–1227.
- Ghannoum M, Gosselin S. Enhanced poison elimination in critical care. *Adv Chronic Kidney Dis*. 2013;20(1):94–101.
- Vale J, Krenzelok EP, Barceloux VD. Position statement and practice guidelines on the use of multi-dose activated charcoal in the treatment of acute poisoning. American Academy of Clinical Toxicology; European Association of Poisons Centres and Clinical Toxicologists. *J Toxicol Clin Toxicol*. 1999;37(6):731–751.
- Proudfoot AT, Krenzelok EP, Vale JA. Position paper on urine alkalization. *J Toxicol Clin Toxicol*. 2004;42(1):1–26.
- Ghannoum M, Nolin TD, Goldfarb DS, Roberts DM, Mactier R, Mowry JB, et al. Extracorporeal treatment for thallium poisoning: recommendations from the EXTRIP Workgroup. *Clin J Am Soc Nephrol*. 2012;7(10):1682–1690.
- Mactier R, Laliberte M, Mardini J, Ghannoum M, Laverne V, Gosselin S, et al. Extracorporeal treatment for barbiturate poisoning: recommendations from the EXTRIP Workgroup. *Am J Kidney Dis*. 2014;64(3):347–358.
- Yates C, Galvao T, Sowinski KM, Mardini K, Botnaru T, Gosselin S, et al. Extracorporeal treatment for tricyclic antidepressant poisoning: recommendations from the EXTRIP Workgroup. *Semin Dial*. 2014;27(4):381–389.
- Ghannoum M, Yates C, Galvao TF, Sowinski KM, Vo TH, Coogan A, et al. Extracorporeal treatment for carbamazepine poisoning: systematic review and recommendations from the EXTRIP workgroup. *Clin Toxicol*. 2014;52(10):993–1004.
- Gosselin S, Juurlink DN, Kielstein JT, Ghannoum M, Laverne V, Nolin TD, et al. Extracorporeal treatment for acetaminophen poisoning: recommendations from the EXTRIP workgroup. *Clin Toxicol*. 2014;52(8):856–867.
- Juurlink DN, Gosselin S, Kielstein JT, Ghannoum M, Laverne V, Nolin TD, et al. Extracorporeal treatment for salicylate poisoning: systematic review and recommendations from the EXTRIP workgroup. *Ann Emerg Med*. 2015;66(2):165–181.
- Ghannoum M, Laliberte M, Nolin TD, MacTier R, Laverne V, Hoffman RS, et al. Extracorporeal treatment for valproic acid poisoning: systematic review and recommendations from the EXTRIP workgroup. *Clin Toxicol*. 2015;53(5):454–465.
- Ghannoum M, Wiegand TJ, Liu KD, Calello DP, Godin M, Laverne V, et al. Extracorporeal treatment for theophylline poisoning: systematic review and recommendations from the EXTRIP workgroup. *Clin Toxicol*. 2015;53(4):215–229.
- Roberts DM, Yates C, Megarbane B, Winchester JF, Maclaren R, Gosselin S, et al. Recommendations for the role of extracorporeal treatments in the management of acute methanol poisoning: a systematic review and consensus statement. *Crit Care Med*. 2015;43(2):461–472.
- Decker BS, Goldfarb DS, Dargan PI, Friesen M, Gosselin S, Hoffman RS, et al. Extracorporeal treatment for lithium poisoning: systematic review and recommendations from the EXTRIP workgroup. *Clin J Am Soc Nephrol*. 2015;10(5):875–887.
- Anseuw K, Mowry JB, Burdmann EA, Ghannoum M, Hoffman RS, Gosselin S, et al. Extracorporeal treatment in phenytoin poisoning: systematic review and recommendations from the EXTRIP (extracorporeal treatments in poisoning) workgroup. *Am J Kidney Dis*. 2016;67(2):187–197.
- Mowry JB, Burdmann EA, Anseuw K, Ayoub P, Ghannoum M, Hoffman RS, et al. Extracorporeal treatment for digoxin poisoning: systematic review and recommendations from the EXTRIP Workgroup. *Clin Toxicol*. 2016;54(2):103–114.
- Ghannoum M, Gosselin S, Hoffman RS, Laverne V, Megarbane B, Hassanian-Moghaddam H, et al. Extracorporeal treatment for ethylene glycol poisoning: systematic review and recommendations from the EXTRIP workgroup. *Crit Care*. 2023;27(1):56.
- Ghannoum M, Roberts DM, Goldfarb DS, Heldrup J, Anseuw K, Galvao TF, et al. Extracorporeal treatment for methotrexate poisoning: systematic review and recommendations from the EXTRIP workgroup. *Clin J Am Soc Nephrol*. 2022;17(4):602–622.
- Bouchard J, Yates C, Calello DP, Gosselin S, Roberts DM, Laverne V, et al. Extracorporeal treatment for gabapentin and pregabalin poisoning: systematic review and recommendations from the EXTRIP workgroup. *Am J Kidney Dis*. 2022;79(1):88–104.
- Bouchard J, Shepherd G, Hoffman RS, Gosselin S, Roberts DM, Li Y, et al. Extracorporeal treatment for poisoning to beta-adrenergic antagonists: systematic review and recommendations from the EXTRIP workgroup. *Crit Care*. 2021;25(1):201.
- Mowry JB, Shepherd G, Hoffman RS, Laverne V, Gosselin S, Nolin TD, et al. Extracorporeal treatments for isoniazid poisoning: systematic review and recommendations from the EXTRIP workgroup. *Pharmacotherapy*. 2021;41(5):463–478.
- Wong A, Hoffman RS, Walsh SJ, Roberts DM, Gosselin S, Bunchman TE, et al. Extracorporeal treatment for calcium channel blocker poisoning: systematic review and recommendations from the EXTRIP workgroup. *Clin Toxicol*. 2021;59(5):361–375.
- Berling I, King JD, Shepherd G, Hoffman RS, Alhatali B, Laverne V, et al. Extracorporeal treatment for chloroquine, hydroxychloroquine, and quinine poisoning: systematic review and recommendations from the EXTRIP workgroup. *J Am Soc Nephrol*. 2020;31(10):2475–2489.
- Ghannoum M, Berling I, Laverne V, Roberts DM, Galvao T, Hoffman RS, et al. Recommendations from the EXTRIP workgroup on extracorporeal treatment for baclofen poisoning. *Kidney Int*. 2021;100(4):720–736.
- Roberts DM, Gallapathy G, Dunuwille A, Chan BS. Pharmacological treatment of cardiac glycoside poisoning. *Br J Clin Pharmacol*. 2016;81(3):488–495.
- Skinner CG, Chang AS, Matthews AS, Reedy SJ, Morgan BW. Randomized controlled study on the use of multiple-dose activated charcoal in patients with supratherapeutic phenytoin levels. *Clin Toxicol*. 2012;50(8):764–769.
- Ghannoum M, Laverne V, Yue CS, Ayoub P, Perreault MM, Roy L. Successful treatment of lithium toxicity with sodium polystyrene sulfonate: a retrospective cohort study. *Clin Toxicol*. 2010;48(1):34–41.
- Hoffman RS. Thallium toxicity and the role of Prussian blue in therapy. *Toxicol Rev*. 2003;22(1):29–40.
- Maduell F, Navarro V, Cruz MC, et al. Osteocalcin and myoglobin removal in on-line hemodiafiltration versus low- and high-flux hemodialysis. *Am J Kidney Dis*. 2002;40(3):582–589.
- Matzke GR, Frye RF. Drug administration in patients with renal insufficiency. Minimising renal and extrarenal toxicity. *Drug Saf*. 1997;16(3):205–231.
- Roberts DM, Dawson AH, Senarathna L, et al. Toxicokinetics, including saturable protein binding, of 4-chloro-2-methyl phenoxyacetic acid (MCPA) in patients with acute poisoning. *Toxicol Lett*. 2011;201(3):270–276.
- Schreiner GE. The role of hemodialysis (artificial kidney) in acute poisoning. *AMA Arch Intern Med*. 1958;102(6):896–913.
- Ozayar E, Degerli S, Gulec H. Hemodiafiltration: a novel approach for treating severe amitriptyline intoxication. *Toxicol Int*. 2012;19(3):319–321.
- Gleeson GH, Nicotero JR, Duffy FC, Kennedy FB, Ticzon AR. Resin hemoperfusion in acute digoxin intoxication. *JAMA*. 1978;240(25):2731.
- Laverne V, Ouellet G, Bouchard J, Galvao T, Kielstein JT, Roberts DM, et al. Guidelines for reporting case studies on extracorporeal treatments in poisonings: methodology. *Semin Dial*. 2014;27(4):407–414.
- Ouellet G, Bouchard J, Ghannoum M, Decker BS. Available extracorporeal treatments for poisoning: overview and limitations. *Semin Dial*. 2014;27(4):342–349.
- Bouchard J, Roberts DM, Roy L, Ouellet G, Decker BS, Mueller BA, et al. Principles and operational parameters to optimize poison removal with extracorporeal treatments. *Semin Dial*. 2014;27(4):371–380.
- Ghannoum M, Roberts DM, Hoffman RS, Ouellet G, Roy L, Decker BS, et al. A stepwise approach for the management of poisoning with extracorporeal treatments. *Semin Dial*. 2014;27(4):362–370.
- Bouchard J, Laverne V, Roberts DM, Cormier M, Morissette G, Ghannoum M. Availability and cost of extracorporeal treatments

- for poisonings and other emergency indications: a worldwide survey. *Nephrol Dial Transplant*. 2017;32(4):699–706.
42. Ghannoum M, Bouchard J, Nolin TD, Ouellet G, Roberts DM. Hemoperfusion for the treatment of poisoning: technology, determinants of poison clearance, and application in clinical practice. *Semin Dial*. 2014;27(4):350–361.
 43. Rahman MH, Haqqie SS, McGoldrick MD. Acute hemolysis with acute renal failure in a patient with valproic acid poisoning treated with charcoal hemoperfusion. *Hemodial Int*. 2006;10(3):256–259.
 44. Shannon MW. Comparative efficacy of hemodialysis and hemoperfusion in severe theophylline intoxication. *Acad Emerg Med*. 1997;4(7):674–678.
 45. Falkenhagen D, Gottschall S, Esther G, Courtney JM, Klinkmann H. In vitro assessment of charcoal and resin hemoabsorbents. *Contrib Nephrol*. 1982;29:23–33.
 46. Mydlik M, Derzsiova K, Bucek J, Horky K, Jarcuska J, Takac M. Use of charcoal haemoperfusion in 55 acute poisonings. *Life Support Syst*. 1983;1(Suppl 1):53–56.
 47. Tapolyai M, Campbell M, Dailey K, Udvari-Nagy S. Hemodialysis is as effective as hemoperfusion for drug removal in carbamazepine poisoning. *Nephron*. 2002;90(2):213–215.
 48. Breuer TKG, Quast DR, Wiciok S, Labedi A, Ellrichmann G. Successful treatment of severe digitoxin intoxication with CytoSorb(R) hemoabsorption. *Blood Purif*. 2021;50(1):137–140.
 49. De Schryver N, Hantson P, Haufroid V, Dechamps M. Cardiogenic shock in a hemodialyzed patient on flecainide: treatment with intravenous fat emulsion, extracorporeal cardiac life support, and CytoSorb(R) hemoabsorption. *Case Rep Cardiol*. 2019;2019:1905871.
 50. Giuntoli L, Dalmastri V, Cilloni N, et al. Severe quetiapine voluntary overdose successfully treated with a new hemoperfusion sorbent. *Int J Artif Organs*. 2019;42(9):516–520.
 51. Kim Z, Goldfarb DS. Continuous renal replacement therapy does not have a clear role in the treatment of poisoning. *Nephron Clin Pract*. 2010;115(1):c1–c6.
 52. Berbece AN, Richardson RM. Sustained low-efficiency dialysis in the ICU: cost, anticoagulation, and solute removal. *Kidney Int*. 2006;70(5):963–968.
 53. Evanson JA, Himmelfarb J, Wingard R, et al. Prescribed versus delivered dialysis in acute renal failure patients. *Am J Kidney Dis*. 1998;32(5):731–738.
 54. Lund B, Seifert SA, Mayersohn M. Efficacy of sustained low-efficiency dialysis in the treatment of salicylate toxicity. *Nephrol Dial Transplant*. 2005;20(7):1483–1484.
 55. Kaplan AA, Bailey RA, Kew CE, Reardon J, Sevigny J. High flux plasma exchange using a modified rotating membrane system. *ASAIO J*. 1996;42(6):957–960.
 56. Choi JH, Oh JC, Kim KH, Chong SY, Kang MS, Oh DY. Successful treatment of cisplatin overdose with plasma exchange. *Yonsei Med J*. 2002;43(1):128–132.
 57. Jha S, Waghdhare S, Reddi R, Bhattacharya P. Thyroid storm due to inappropriate administration of a compounded thyroid hormone preparation successfully treated with plasmapheresis. *Thyroid*. 2012;22(12):1283–1286.
 58. Pierga JY, Beuzeboc P, Dorval T, Palangie T, Poullart P. Favourable outcome after plasmapheresis for vincristine overdose. *Lancet*. 1992;340(8812):185.
 59. Hastings D, Patel B, Torloni AS, et al. Plasmapheresis therapy for rare but potentially fatal reaction to rituximab. *J Clin Apher*. 2009;24(1):28–31.
 60. Couriel D, Weinstein R. Complications of therapeutic plasma exchange: a recent assessment. *J Clin Apher*. 1994;9(1):1–5.
 61. Perino GC, Grivet V. Hemoperfusion and plasmapheresis complications. *Minerva Urol Nefrol*. 1987;39(2):161–163.
 62. Kantola T, Koivusalo AM, Hockerstedt K, Isoniemi H. Early molecular adsorbents recirculating system treatment of Amanita mushroom poisoning. *Ther Apher Dial*. 2009;13(5):399–403.
 63. Sein Anand J, Chodorowski Z, Hydzik P. Molecular adsorbent recirculating system–MARS as a bridge to liver transplantation in Amanita phalloides intoxication. *Przegl Lek*. 2005;62(6):480–481.
 64. Lione C, Sorodoc L, Simionescu V. Successful treatment of an adult with Amanita phalloides-induced fulminant liver failure with molecular adsorbent recirculating system (MARS). *Rom J Gastroenterol*. 2005;14(3):267–271.
 65. Hydzik P, Drozd M, Sulowicz W, Groszek B. Liver albumin dialysis–application in acetaminophen poisoning. *Przegl Lek*. 2004;61(4):377–382.
 66. Dichtwald S, Dahan E, Adi N, Moses A, Sorkine P. Molecular adsorbent recycling system therapy in the treatment of acute valproic acid intoxication. *Isr Med Assoc J*. 2010;12(5):307–308.
 67. Sen S, Ratnaraj N, Davies NA, et al. Treatment of phenytoin toxicity by the molecular adsorbents recirculating system (MARS). *Epilepsia*. 2003;44(2):265–267.
 68. Korseed S, Selby NM, Fluck RJ. Treatment of severe theophylline poisoning with the molecular adsorbent recirculating system (MARS). *Nephrol Dial Transplant*. 2007;22(3):969–970.
 69. Ghannoum M, Nolin TD, Lavergne V, Hoffman RS. Blood purification in toxicology: nephrology's ugly duckling. *Adv Chronic Kidney Dis*. 2011;18(3):160–166.
 70. Lavergne V, Nolin TD, Hoffman RS, Robert D, Gosselin S, Goldfarb DS, et al. The EXTRIP (extracorporeal treatments in poisoning) workgroup: guideline methodology. *Clin Toxicol*. 2012;50:403–413.
 71. Calello DP, Liu KD, Wiegand TJ, Roberts DM, Lavergne V, Gosselin S, et al. Extracorporeal treatment for metformin poisoning: systematic review and recommendations from the extracorporeal treatments in poisoning workgroup. *Crit Care Med*. 2015;43(8):1716–1730.
 72. Ellsworth H, Engebretsen KM, Hlavenka LM, et al. A cost comparison of fomepizole and hemodialysis in the treatment of methanol and ethylene glycol toxicity. *Clin Toxicol*. 2011;49:590–591.
 73. Rabindranath KS, Kumar E, Shail R, Vaux EC. Ultrasound use for the placement of haemodialysis catheters. *Cochrane Database Syst Rev*. 2011;(11):CD005279.
 74. Ghannoum M, Kazim S, Grunbaum AM, Villeneuve E, Gosselin S. Massive acetaminophen overdose: effect of hemodialysis on acetaminophen and acetylcysteine kinetics. *Clin Toxicol*. 2016;54(6):519–522.
 75. Prabhu MV, Juneja D, Gopal PB, et al. Ultrasound-guided femoral dialysis access placement: a single-center randomized trial. *Clin J Am Soc Nephrol*. 2010;5(2):235–239.
 76. Leblanc M, Fedak S, Mokris G, Paganini EP. Blood recirculation in temporary central catheters for acute hemodialysis. *Clin Nephrol*. 1996;45(5):315–319.
 77. Parienti JJ, Megarbane B, Fischer MO, et al. Catheter dysfunction and dialysis performance according to vascular access among 736 critically ill adults requiring renal replacement therapy: a randomized controlled study. *Crit Care Med*. 2010;38(4):1118–1125.
 78. Panzer U, Kluge S, Kreymann G, Wolf G. Combination of intermittent haemodialysis and high-volume continuous haemofiltration for the treatment of severe metformin-induced lactic acidosis. *Nephrol Dial Transplant*. 2004;19(8):2157–2158.
 79. Friesseck S, Abel P, Kraft M, Gerner A, Runge S. Combined renal replacement therapy for severe metformin-induced lactic acidosis. *Nephrol Dial Transplant*. 2006;21(7):2038–2039.
 80. Fertel BS, Nelson LS, Goldfarb DS. The underutilization of hemodialysis in patients with salicylate poisoning. *Kidney Int*. 2009;75(12):1349–1353.
 81. Hansen HE, Amdisen A. Lithium intoxication. (Report of 23 cases and review of 100 cases from the literature). *Q J Med*. 1978;47(186):123–144.
 82. Bouchard J, Ghannoum M, Bernier-Jean A, et al. Comparison of intermittent and continuous extracorporeal treatments for the enhanced elimination of dabigatran. *Clin Toxicol*. 2015;53(3):156–163.
 83. Grimes DJ, Bowles MR, Buttsworth JA, et al. Survival after unexpected high serum methotrexate concentrations in a patient with osteogenic sarcoma. *Drug Saf*. 1990;5(6):447–454.
 84. Amdisen A, Skjoldborg H. Haemodialysis for lithium poisoning. *Lancet*. 1969;2(7613):213.
 85. Treysman L, Meehan TJ, Schlieben DJ, Ducre B, Erickson TB. Pharmacokinetic modeling of lithium elimination during 67 continuous hours of high flux hemodialysis. *Clin Toxicol*. 2010;48(6):647.
 86. Lavergne V, Hoffman RS, Mowry JB, Cormier M, Gosselin S, Roberts DM, et al. Why are we still dialyzing overdoses to tricyclic antidepressants? A subanalysis of the npds database. *Semin Dial*. 2016;29(5):403–409.

87. Jacobsen D, Ovrebo S, Ostborg J, Sejersted OM. Glycolate causes the acidosis in ethylene glycol poisoning and is effectively removed by hemodialysis. *Acta Med Scand*. 1984;216(4):409–416.
88. Keyhani J, Keyhani E. EPR study of the effect of formate on cytochrome C oxidase. *Biochem Biophys Res Commun*. 1980;92(1):327–333.
89. Abramson S, Singh AK. Treatment of the alcohol intoxications: ethylene glycol, methanol and isopropanol. *Curr Opin Nephrol Hypertens*. 2000;9(6):695–701.
90. Daniel DR, McAnalley BH, Garriott JC. Isopropyl alcohol metabolism after acute intoxication in humans. *J Anal Toxicol*. 1981;5(3):110–112.
91. Gaudet MP, Fraser GL. Isopropanol ingestion: case report with pharmacokinetic analysis. *Am J Emerg Med*. 1989;7(3):297–299.
92. Natowicz M, Donahue J, Gorman L, Kane M, McKissick J, Shaw L. Pharmacokinetic analysis of a case of isopropanol intoxication. *Clin Chem*. 1985;31(2):326–328.
93. Jones AW. Elimination half-life of acetone in humans: case reports and review of the literature. *J Anal Toxicol*. 2000;24(1):8–10.
94. Parker KM, Lera Jr TA. Acute isopropanol ingestion: pharmacokinetic parameters in the infant. *Am J Emerg Med*. 1992;10(6):542–544.
95. Zaman F, Pervez A, Abreo K. Isopropyl alcohol intoxication: a diagnostic challenge. *Am J Kidney Dis*. 2002;40(3):E12.
96. Sharma R, Marasini S, Sharma AK, Shrestha JK, Nepal BP. Methanol poisoning: ocular and neurological manifestations. *Optom Vis Sci*. 2012;89(2):178–182.
97. Lin SH, Tseng GF, Liang CC, Hung YC. Methanol intoxication with bilateral putaminal and occipital necrosis. *Tzu Chi Med J*. 2010;22(3):160–163.
98. Church AS, Witting MD. Laboratory testing in ethanol, methanol, ethylene glycol, and isopropanol toxicities. *J Emerg Med*. 1997;15(5):687–692.
99. Barceloux DG, Krenzelok EP, Olson K, Watson W. American academy of clinical toxicology practice guidelines on the treatment of ethylene glycol poisoning. Ad hoc committee. *J Toxicol Clin Toxicol*. 1999;37(5):537–560.
100. Hoffman RS, Smilkstein MJ, Howland MA, Goldfrank LR. Osmol gaps revisited: normal values and limitations. *J Toxicol Clin Toxicol*. 1993;31(1):81–93.
101. Pursell RA, Lynd LD, Koga Y. The use of the osmole gap as a screening test for the presence of exogenous substances. *Toxicol Rev*. 2004;23(3):189–202.
102. Monaghan MS, Ackerman BH, Olsen KM, Farmer C, Pappas AA. The use of delta osmolality to predict serum isopropanol and acetone concentrations. *Pharmacotherapy*. 1993;13(1):60–63.
103. Glaser DS. Utility of the serum osmol gap in the diagnosis of methanol or ethylene glycol ingestion. *Ann Emerg Med*. 1996;27(3):343–346.
104. Hovda KE, Hunderi OH, Rudberg N, Froyshov S, Jacobsen D. Anion and osmolal gaps in the diagnosis of methanol poisoning: clinical study in 28 patients. *Intensive Care Med*. 2004;30(9):1842–1846.
105. Roberts DM, Hoffman RS, Brent J, Laverne V, Hovda KE, Porter WH, et al. The serum glycolate concentration: its prognostic value and its correlation to surrogate markers in ethylene glycol exposures. *Clin Toxicol*. 2022;60(7):798–807.
106. Christiansson LK, Kaspersson KE, Kulling PE, Ovrebo S. Treatment of severe ethylene glycol intoxication with continuous arteriovenous hemofiltration dialysis. *J Toxicol Clin Toxicol*. 1995;33(3):267–270.
107. Jacobsen D, Hewlett TP, Webb R, Brown ST, Ordinario AT, McMartin KE. Ethylene glycol intoxication: evaluation of kinetics and crystalluria. *Am J Med*. 1988;84(1):145–152.
108. Casavant MJ, Shah MN, Battels R. Does fluorescent urine indicate antifreeze ingestion by children? *Pediatrics*. 2001;107(1):113–114.
109. Sharma AN, O'Shaughnessy PM, Hoffman RS. Urine fluorescence: is it a good test for ethylene glycol ingestion? *Pediatrics*. 2002;109(2):345.
110. Barceloux DG, Bond GR, Krenzelok EP, Cooper H, Vale JA. American academy of clinical toxicology practice guidelines on the treatment of methanol poisoning. *J Toxicol Clin Toxicol*. 2002;40(4):415–446.
111. Su M, Hoffman RS, Nelson LS. Error in an emergency medicine textbook: isopropyl alcohol toxicity. *Acad Emerg Med*. 2002;9(2):175.
112. Megarbane B, Baud F. What is the role of bicarbonate in the management of acidosis in the poisoned patient. *Clin Toxicol*. 2006;44(4):415–416.
113. Beaulieu J, Roberts DM, Gosselin S, Hoffman RS, Laverne V, Hovda KE, et al. Treating ethylene glycol poisoning with alcohol dehydrogenase inhibition, but without extracorporeal treatments: a systematic review. *Clin Toxicol*. 2022;60(7):784–797.
114. Megarbane B, Baud F. Is there a remaining place for hemodialysis in toxic alcohol poisonings treated with fomepizole? *Clin Toxicol*. 2003;41(4):396–397.
115. Burns MJ, Graudins A, Aaron CK, McMartin K, Brent J. Treatment of methanol poisoning with intravenous 4-methylpyrazole. *Ann Emerg Med*. 1997;30(6):829–832.
116. Darracq MA, Rentmeester LL, Clark RF, Tomaszewski CA, Schneir AB, Cantrell FL. Cost of hemodialysis versus fomepizole-only for treatment of ethylene glycol intoxication. *Clin Toxicol*. 2013;51(3):188.
117. Iliuta IA, Lachance P, Ghannoum M, et al. Prediction and validation of the duration of hemodialysis sessions for the treatment of acute ethylene glycol poisoning. *Kidney Int*. 2017;92(2):453–460.
118. Lachance P, Mac-Way F, Desmeules S, et al. Prediction and validation of hemodialysis duration in acute methanol poisoning. *Kidney Int*. 2015;88(5):1170–1177.
119. Roberts DM, Lea-Henry TN. Simplifying the hemodialysis prescription in patients with ethylene glycol poisoning. *Kidney Int*. 2017;92(2):291–293.
120. Garella S. Extracorporeal techniques in the treatment of exogenous intoxications. *Kidney Int*. 1988;33(3):735–754.
121. Lacouture PG, Wason S, Abrams A, Lovejoy Jr FH. Acute isopropyl alcohol intoxication. Diagnosis and management. *Am J Med*. 1983;75(4):680–686.
122. Freireich AW, Cinque TJ, Xanthaky G, Landau D. Hemodialysis for isopropanol poisoning. *N Engl J Med*. 1967;277(13):699–700.
123. Mason L, Moore RA, Edwards JE, McQuay HJ, Derry S, Wiffen PJ. Systematic review of efficacy of topical rubefacients containing salicylates for the treatment of acute and chronic pain. *BMJ*. 2004;328(7446):995.
124. Tramer MR. It's not just about rubbing—topical capsaicin and topical salicylates may be useful as adjuvants to conventional pain treatment. *BMJ*. 2004;328(7446):998.
125. Sogge MR, Griffith JL, Sinar DR, Mayes GR. Lavage to remove enteric-coated aspirin and gastric outlet obstruction. *Ann Intern Med*. 1977;87(6):721–722.
126. Wortzman DJ, Grunfeld A. Delayed absorption following enteric-coated aspirin overdose. *Ann Emerg Med*. 1987;16(4):434–436.
127. Levy G. Clinical pharmacokinetics of salicylates: a re-assessment. *Br J Clin Pharmacol*. 1980;10(suppl 2):285S–90S.
128. Levy G. Pharmacokinetics of salicylate elimination in man. *J Pharm Sci*. 1965;54(7):959–967.
129. Swintosky JV. Illustrations and pharmaceutical interpretations of first order drug elimination rate from the bloodstream. *J Am Pharm Assoc*. 1956;45(6):395–400.
130. Alvan G, Bergman U, Gustafsson LL. High unbound fraction of salicylate in plasma during intoxication. *Br J Clin Pharmacol*. 1981;11(6):625–626.
131. Borga O, Cederlof IO, Ringberger VA, Norlin A. Protein binding of salicylate in uremic and normal plasma. *Clin Pharmacol Ther*. 1976;20(4):464–475.
132. Hill JB. Experimental salicylate poisoning: observations on the effects of altering blood pH on tissue and plasma salicylate concentrations. *Pediatrics*. 1971;47(4):658–665.
133. Gabow PA, Anderson RJ, Potts DE, Schrier RW. Acid-base disturbances in the salicylate-intoxicated adult. *Arch Intern Med*. 1978;138(10):1481–1484.
134. Tenney SM, Miller RM. The respiratory and circulatory actions of salicylate. *Am J Med*. 1955;19(4):498–508.
135. Winters RW, White JS, Hughes MC, Ordway NK. Disturbances of acid-base equilibrium in salicylate intoxication. *Pediatrics*. 1959;23(2):260–285.
136. Yip L, Dart RC, Gabow PA. Concepts and controversies in salicylate toxicity. *Emerg Med Clin North Am*. 1994;12(2):351–364.
137. Chyka PA, Erdman AR, Christianson G, et al. Salicylate poisoning: an evidence-based consensus guideline for out-of-hospital management. *Clin Toxicol*. 2007;45(2):95–131.

138. Temple AR. Acute and chronic effects of aspirin toxicity and their treatment. *Arch Intern Med.* 1981;141(3):364–369.
139. Proudfoot AT, Brown SS. Acidaemia and salicylate poisoning in adults. *Br Med J.* 1969;2(5656):547–550.
140. Raschke R, Arnold-Capell PA, Richeson R, Curry SC. Refractory hypoglycemia secondary to topical salicylate intoxication. *Arch Intern Med.* 1991;151(3):591–593.
141. Anderson RJ, Potts DE, Gabow PA, Rumack BH, Schrier RW. Unrecognized adult salicylate intoxication. *Ann Intern Med.* 1976;85(6):745–748.
142. Bailey RB, Jones SR. Chronic salicylate intoxication. A common cause of morbidity in the elderly. *J Am Geriatr Soc.* 1989;37(6):556–561.
143. Walters JS, Woodring JH, Stelling CB, Rosenbaum HD. Salicylate-induced pulmonary edema. *Radiology.* 1983;146(2):289–293.
144. Fisher Jr CJ, Albertson TE, Foulke GE. Salicylate-induced pulmonary edema: clinical characteristics in children. *Am J Emerg Med.* 1985;3(1):33–37.
145. Done AK, Otterness LJ. Exchange transfusion in the treatment of oil of wintergreen (methyl salicylate) poisoning. *Pediatrics.* 1956;18(1):80–85.
146. Greenberg MI, Hendrickson RG, Hofman M. Deleterious effects of endotracheal intubation in salicylate poisoning. *Ann Emerg Med.* 2003;41(4):583–584.
147. Neuvonen PJ, Elfving SM, Elonen E. Reduction of absorption of digoxin, phenytoin and aspirin by activated charcoal in man. *Eur J Clin Pharmacol.* 1978;13(3):213–218.
148. Levy G, Tsuchiya T. Effect of activated charcoal on aspirin absorption in man. Part I. *Clin Pharmacol Ther.* 1972;13(3):317–322.
149. Kirshenbaum LA, Mathews SC, Sitar DS, Tenenbein M. Whole-bowel irrigation versus activated charcoal in sorbitol for the ingestion of modified-release pharmaceuticals. *Clin Pharmacol Ther.* 1989;46(3):264–271.
150. Barone JA, Raia JJ, Huang YC. Evaluation of the effects of multiple-dose activated charcoal on the absorption of orally administered salicylate in a simulated toxic ingestion model. *Ann Emerg Med.* 1988;17(1):34–37.
151. Prescott LF, Balali-Mood M, Critchley JA, Johnstone AF, Proudfoot AT. Diuresis or urinary alkalisation for salicylate poisoning? *Br Med J.* 1982;285(6352):1383–1386.
152. Proudfoot AT, Krenzelok EP, Brent J, Vale JA. Does urine alkalization increase salicylate elimination? If so, why? *Toxicol Rev.* 2003;22(3):129–136.
153. Vree TB, Van Ewijk-Beneken Kolmer EW, Verwey-Van Wissen CP, Hekster YA. Effect of urinary pH on the pharmacokinetics of salicylic acid, with its glycine and glucuronide conjugates in human. *Int J Clin Pharmacol Ther.* 1994;32(10):550–558.
154. Hill JB, Palaia FL, McAdams JL, Palmer PJ, Maret SM. Efficacy of activated charcoal hemoperfusion in removing lethal doses of barbiturates and salicylate from the blood of rats and dogs. *Clin Chem.* 1976;22(6):754–760.
155. James JA, Kimbell L, Read WT. Experimental salicylate intoxication. I. Comparison of exchange transfusion, intermittent peritoneal lavage, and hemodialysis as means for removing salicylate. *Pediatrics.* 1962;29:442–447.
156. Widdop B, Medd RK, Braithwaite RA, Rees AJ, Goulding R. Experimental drug intoxication: treatment with charcoal haemoperfusion. *Arch Toxicol.* 1975;34(1):27–36.
157. Wrathall G, Sinclair R, Moore A, Pogson D. Three case reports of the use of haemodiafiltration in the treatment of salicylate overdose. *Hum Exp Toxicol.* 2001;20(9):491–495.
158. French LK, McKeown NJ, Hendrickson RG. Continuous renal replacement therapy for salicylate overdose in a patient with multi-system trauma. *Clin Toxicol.* 2011;49:541–542.
159. Chase PB, Walter FG, James S. Whole bowel irrigation, hemodialysis, and continuous venovenous hemodiafiltration in the successful treatment of severe salicylate poisoning: case report. *Dial Transplant.* 2002;31(6):387+90–92+419.
160. Dargan PI, Wallace CI, Jones AL. An evidence based flowchart to guide the management of acute salicylate (aspirin) overdose. *Emerg Med J.* 2002;19(3):206–209.
161. McKnight RF, Adida M, Budge K, Stockton S, Goodwin GM, Geddes JR. Lithium toxicity profile: a systematic review and meta-analysis. *Lancet.* 2012;379(9817):721–728.
162. Einat H, Kofman O, Itkin O, Lewitan RJ, Belmaker RH. Augmentation of lithium's behavioral effect by inositol uptake inhibitors. *J Neural Transm.* 1998;105(1):31–38.
163. Klein PS, Melton DA. A molecular mechanism for the effect of lithium on development. *Proc Natl Acad Sci U S A.* 1996;93(16):8455–8459.
164. Yin L, Wang J, Klein PS, Lazar MA. Nuclear receptor Rev-erb α is a critical lithium-sensitive component of the circadian clock. *Science.* 2006;311(5763):1002–1005.
165. Jope RS. Anti-bipolar therapy: mechanism of action of lithium. *Mol Psychiatry.* 1999;4(2):117–128.
166. Jaeger A, Sauder P, Kopferschmitt J, Tritsch L, Flesch F. When should dialysis be performed in lithium poisoning? A kinetic study in 14 cases of lithium poisoning. *J Toxicol Clin Toxicol.* 1993;31(3):429–447.
167. Timmer RT, Sands JM. Lithium intoxication. *J Am Soc Nephrol.* 1999;10(3):666–674.
168. Finley PR, Warner MD, Peabody CA. Clinical relevance of drug interactions with lithium. *Clin Pharmacokinet.* 1995;29(3):172–191.
169. Baird-Gunning J, Lea-Henry T, Hoegberg LCG, Gosselin S, Roberts DM. Lithium poisoning. *J Intensive Care Med.* 2017;32(4):249–263.
170. Grandjean EM, Aubry JM. Lithium: updated human knowledge using an evidence-based approach: part III: clinical safety. *CNS Drugs.* 2009;23(5):397–418.
171. Speirs J, Hirsch SR. Severe lithium toxicity with “normal” serum concentrations. *Br Med J.* 1978;1(6116):815–816.
172. Tilikian AG, Schroeder JS, Kao JJ, Hultgren HN. The cardiovascular effects of lithium in man. A review of the literature. *Am J Med.* 1976;61(5):665–670.
173. Offerman SR, Alsop JA, Lee J, Holmes JF. Hospitalized lithium overdose cases reported to the California Poison Control System. *Clin Toxicol.* 2010;48(5):443–448.
174. Adityanjee MKR, Thampany A. The syndrome of irreversible lithium-effectuated neurotoxicity. *Clin Neuropharmacol.* 2005;28(1):38–49.
175. Favin FD, Klein-Schwartz W, Oderda GM, Rose SR. In vitro study of lithium carbonate adsorption by activated charcoal. *J Toxicol Clin Toxicol.* 1988;26(7):443–450.
176. Smith SW, Ling LJ, Halstenson CE. Whole-bowel irrigation as a treatment for acute lithium overdose. *Ann Emerg Med.* 1991;20(5):536–539.
177. Linakis JG, Savitt DL, Wu TY, Lockhart GR, Lacouture PG. Use of sodium polystyrene sulfonate for reduction of plasma lithium concentrations after chronic lithium dosing in mice. *J Toxicol Clin Toxicol.* 1998;36(4):309–313.
178. Belanger DR, Tierney MG, Dickinson G. Effect of sodium polystyrene sulfonate on lithium bioavailability. *Ann Emerg Med.* 1992;21(11):1312–1315.
179. Bailey B, McGuigan M. Comparison of patients hemodialyzed for lithium poisoning and those for whom dialysis was recommended by PCC but not done: what lesson can we learn? *Clin Nephrol.* 2000;54(5):388–392.
180. Eyer F, Pfab R, Felgenhauer N, et al. Lithium poisoning: pharmacokinetics and clearance during different therapeutic measures. *J Clin Psychopharmacol.* 2006;26(3):325–330.
181. Leblanc M, Raymond M, Bonnardeaux A, et al. Lithium poisoning treated by high-performance continuous arteriovenous and venovenous hemodiafiltration. *Am J Kidney Dis.* 1996;27(3):365–372.
182. Peces R, Fernandez EJ, Regidor D, et al. Treatment of acute lithium intoxication with high-flux haemodialysis membranes. *Nefrologia.* 2006;26(3):372–378.
183. Schmidt JJ, Lorenzen J, Chatzikyrkou C, Lichtinghagen R, Kielstein JT. Total collected dialysate lithium concentration after successful dialysis treatment in case of intoxication. *BMC Pharmacol Toxicol.* 2014;15:49.
184. Jacobsen D, Aasen G, Frederichsen P, Eisenga B. Lithium intoxication: pharmacokinetics during and after terminated hemodialysis in acute intoxications. *J Toxicol Clin Toxicol.* 1987;25(1–2):81–94.
185. Ayuso Gatell A, Leon Regidor MA, Mestre Saura J, Diaz Boladeras RM, Sirvent Calvera JM, Nolla Panades M. Acute lithium poisoning. Treatment with continuous arteriovenous hemofiltration. Apoptosis of a case. *Rev Clin Esp.* 1989;185(4):195–197.

186. Bellomo R, Kearly Y, Parkin G, Love J, Boyce N. Treatment of life-threatening lithium toxicity with continuous arterio-venous hemodiafiltration. *Crit Care Med*. 1991;19(6):836–837.
187. Meyer RJ, Flynn JT, Brophy PD, et al. Hemodialysis followed by continuous hemofiltration for treatment of lithium intoxication in children. *Am J Kidney Dis*. 2001;37(5):1044–1047.
188. Wilson JH, Donker AJ, van der Hem GK, Wientjes J. Peritoneal dialysis for lithium poisoning. *Br Med J*. 1971;2(764):749–750.
189. Winchester JF. Evolution of artificial organs: extracorporeal removal of drugs. *Artif Organs*. 1986;10(4):316–323.
190. Clendeninn NJ, Pond SM, Kaysen G, Barraza JJ, Farrell T, Becker CE. Potential pitfalls in the evaluation of the usefulness of hemodialysis for the removal of lithium. *J Toxicol Clin Toxicol*. 1982;19(4):341–352.
191. Hicks LK, McFarlane PA. Valproic acid overdose and haemodialysis. *Nephrol Dial Transplant*. 2001;16(7):1483–1486.
192. Isbister GK, Balit CR, Whyte IM, Dawson A. Valproate overdose: a comparative cohort study of self poisonings. *Br J Clin Pharmacol*. 2003;55(4):398–404.
193. Eyer F, Felgenhauer N, Gempel K, Steimer W, Gerbitz KD, Zilker T. Acute valproate poisoning: pharmacokinetics, alteration in fatty acid metabolism, and changes during therapy. *J Clin Psychopharmacol*. 2005;25(4):376–380.
194. Kane SL, Constantiner M, Staubus AE, Meinecke CD, Sedor JR. High-flux hemodialysis without hemoperfusion is effective in acute valproic acid overdose. *Ann Pharmacother*. 2000;34(10):1146–1151.
195. Garnier R, Boudignat O, Fournier PE. Valproate poisoning. *Lancet*. 1982;2(8289):97.
196. Sztajnkrzyer MD. Valproic acid toxicity: overview and management. *J Toxicol Clin Toxicol*. 2002;40(6):789–801.
197. McCall M, Bourgeois JA. Valproic acid-induced hyperammonemia: a case report. *J Clin Psychopharmacol*. 2004;24(5):521–526.
198. Nasa P, Sehrawat D, Kansal S, Chawla R. Effectiveness of hemodialysis in a case of severe valproate overdose. *Indian J Crit Care Med*. 2011;15(2):120–122.
199. Davison AS, Milan AM, Roberts NB. The consequences of valproate overdose. *Clin Chem*. 2011;57(9):1233–1237.
200. Perrott J, Murphy NG, Zed PJ. L-carnitine for acute valproic acid overdose: a systematic review of published cases. *Ann Pharmacother*. 2010;44(7–8):1287–1293.
201. Nguyen P, Chevillard L, Gouda AS, Goullain H, Labat L, Malissin I, et al. L-carnitine does not improve valproic acid poisoning management: a cohort study with toxicokinetics and concentration/effect relationships. *Ann Intensive Care*. 2022;12(1):7.
202. Singh SM, McCormick BB, Mustata S, Thompson M, Prasad GV. Extracorporeal management of valproic acid overdose: a large regional experience. *J Nephrol*. 2004;17(1):43–49.
203. Franssen EJ, van Essen GG, Portman AT, et al. Valproic acid toxicokinetics: serial hemodialysis and hemoperfusion. *Ther Drug Monit*. 1999;21(3):289–292.
204. Goodman JW, Goldfarb DS. The role of continuous renal replacement therapy in the treatment of poisoning. *Semin Dial*. 2006;19(5):402–407.
205. Churchill MD, Pasko DA, Smoyer WE, Mueller BA. Enhanced clearance of highly protein-bound drugs by albumin-supplemented dialysate during modeled continuous hemodialysis. *Nephrol Dial Transplant*. 2009;24(1):231–238.
206. Khan E, Huggan P, Celi L, MacGinley R, Schollum J, Walker R. Sustained low-efficiency dialysis with filtration (SLEDD-f) in the management of acute sodium valproate intoxication. *Hemodial Int*. 2008;12(2):211–214.
207. Graudins A, Peden G, Dowsett RP. Massive overdose with controlled-release carbamazepine resulting in delayed peak serum concentrations and life-threatening toxicity. *Emerg Med*. 2002;14(1):89–94.
208. Vree TB, Janssen TJ, Hekster YA, Termond EF, van de Dries AC, Wijnands WJ. Clinical pharmacokinetics of carbamazepine and its epoxy and hydroxy metabolites in humans after an overdose. *Ther Drug Monit*. 1986;8(3):297–304.
209. Brahmi N, Kouraichi N, Thabet H, Amamou M. Influence of activated charcoal on the pharmacokinetics and the clinical features of carbamazepine poisoning. *Am J Emerg Med*. 2006;24(4):440–443.
210. Mise S, Jukic I, Tonkic A, Titlic M, Mise S. Multidose activated charcoal in the treatment of carbamazepine overdose with seizures: a case report. *Arh Hig Rada Toksikol*. 2005;56(4):333–338.
211. Wason S, Baker RC, Carolan P, Seigel R, Druckenbrod RW. Carbamazepine overdose—the effects of multiple dose activated charcoal. *J Toxicol Clin Toxicol*. 1992;30(1):39–48.
212. Montoya-Cabrera MA, Saucedo-Garcia JM, Escalante-Galindo P, Flores-Alvarez E, Ruiz-Gomez A. Carbamazepine poisoning in adolescent suicide attempters. Effectiveness of multiple-dose activated charcoal in enhancing carbamazepine elimination. *Arch Med Res*. 1996;27(4):485–489.
213. Sikma MA, van den Broek MP, Meulenbelt J. Increased unbound drug fraction in acute carbamazepine intoxication: suitability and effectiveness of high-flux haemodialysis. *Intensive Care Med*. 2012.
214. Kielstein JT, Schwarz A, Arnavaz A, Sehlberg O, Emrich HM, Fliser D. High-flux hemodialysis—an effective alternative to hemoperfusion in the treatment of carbamazepine intoxication. *Clin Nephrol*. 2002;57(6):484–486.
215. Muller CH, Lankisch PG. Resin hemoperfusion for carbamazepine intoxication. *Intensivmed Notfallmed*. 1989;26(3):145–148.
216. Payette A, Ghannoum M, Madore F, Albert M, Troyanov S, Bouchard J. Carbamazepine poisoning treated by multiple extracorporeal treatments. *Clin Nephrol*. 2015;83(3):184–188.
217. Maslov OG, Brusin KM, Kochmashev VF, Sentsov VG. Comparative evaluation of methods of extracorporeal detoxification in acute carbamazepine poisoning. *Klinicheskaya Toksikologiya*. 2011;12:1169–1179.
218. Deshpande G, Meert KL, Valentini RP. Repeat charcoal hemoperfusion treatments in life threatening carbamazepine overdose. *Pediatr Nephrol*. 1999;13(9):775–777.
219. de Groot G, van Heijst AN, Maes RA. Charcoal hemoperfusion in the treatment of two cases of acute carbamazepine poisoning. *J Toxicol Clin Toxicol*. 1984;22(4):349–362.
220. Harder JL, Heung M, Vilay AM, Mueller BA, Segal JH. Carbamazepine and the active epoxide metabolite are effectively cleared by hemodialysis followed by continuous venovenous hemodialysis in an acute overdose. *Hemodial Int*. 2011;15(3):412–415.
221. Duzova A, Baskin E, Usta Y, Ozen S. Carbamazepine poisoning: treatment with plasma exchange. *Hum Exp Toxicol*. 2001;20(4):175–177.
222. Azak A, Kocak G, Huddam B, Duranay M. Is conventional hemodialysis enough to manage carbamazepine intoxication? *Blood Purif*. 2012;33(4):225–226.
223. Roberts DM, Buckley NA. Enhanced elimination in acute barbiturate poisoning - a systematic review. *Clin Toxicol*. 2011;49(1):2–12.
224. Jacobs F, Brivet FG. Conventional haemodialysis significantly lowers toxic levels of phenobarbital. *Nephrol Dial Transplant*. 2004;19(6):1663–1664.
225. Patil VC, Patil HV, Sakaria A. Accelerated elimination with charcoal hemoperfusion in acute phenobarbital intoxication: a case report. *Med Leg Update*. 2011;11(2):112–113.
226. Berman LB, Jeghers HJ, Schreiner GE, Pallotta AJ. Hemodialysis, an effective therapy for acute barbiturate poisoning. *J Am Med Assoc*. 1956;161(9):820–827.
227. Ferguson MJ, Grace WJ. The conservative management of barbiturate intoxication: experience with 95 unconscious patients. *Ann Intern Med*. 1961;54:726–733.
228. Setter JG, Maher JF, Schreiner GE. Barbiturate intoxication. Evaluation of therapy including dialysis in a large series selectively referred because of severity. *Arch Intern Med*. 1966;117(2):224–236.
229. Neuvonen PJ, Elonen E. Effect of activated charcoal on absorption and elimination of phenobarbital, carbamazepine and phenylbutazone in man. *Eur J Clin Pharmacol*. 1980;17(1):51–57.
230. Pond SM, Olson KR, Osterloh JD, Tong TG. Randomized study of the treatment of phenobarbital overdose with repeated doses of activated charcoal. *JAMA*. 1984;251(23):3104–3108.
231. Mohammed Ebid AH, Abdel-Rahman HM. Pharmacokinetics of phenobarbital during certain enhanced elimination modalities to evaluate their clinical efficacy in management of drug overdose. *Ther Drug Monit*. 2001;23(3):209–216.
232. Linton AL, Luke RG, Speirs I, Kennedy AC. Forced diuresis and haemodialysis in severe barbiturate intoxication. *Lancet*. 1964;1(7341):1008–1010.
233. Frenia ML, Schauben JL, Wears RL, Karlix JL, Tucker CA, Kuni-saki TA. Multiple-dose activated charcoal compared to urinary

- alkalinization for the enhancement of phenobarbital elimination. *J Toxicol Clin Toxicol*. 1996;34(2):169–175.
234. Hampel G, Wiseman H, Widdop B. Acute poisoning due to hypnotics: the role of haemoperfusion in clinical perspective. *Vet Hum Toxicol*. 1979;21(Suppl):4–6.
 235. Chaikin P, Adir J. Unusual absorption profile of phenytoin in a massive overdose case. *J Clin Pharmacol*. 1987;27(1):70–73.
 236. Hvidberg EF, Dam M. Clinical pharmacokinetics of anticonvulsants. *Clin Pharmacokinet*. 1976;1(3):161–188.
 237. Craig S. Phenytoin poisoning. *Neurocrit Care*. 2005;3(2):161–170.
 238. Oellerich M, Muller-Vahl H. The EMIT FreeLevel ultrafiltration technique compared with equilibrium dialysis and ultracentrifugation to determine protein binding of phenytoin. *Clin Pharmacokinet*. 1984;9(Suppl 1):61–70.
 239. Mauro LS, Mauro VF, Brown DL, Somani P. Enhancement of phenytoin elimination by multiple-dose activated charcoal. *Ann Emerg Med*. 1987;16(10):1132–1135.
 240. Howard CE, Roberts RS, Ely DS, Moye RA. Use of multiple-dose activated charcoal in phenytoin toxicity. *Ann Pharmacother*. 1994;28(2):201–203.
 241. Baehler RW, Work J, Smith W, Dominic JA. Charcoal hemoperfusion in the therapy for methsuximide and phenytoin overdose. *Arch Intern Med*. 1980;140(11):1466–1468.
 242. Kanayama Y, Itakura Y, Iwasaki M, et al. Changes in phenytoin concentrations in blood and cerebrospinal fluid caused by direct hemoperfusion in a patient intoxicated with phenytoin. *Ther Apher*. 1998;2(1):74–77.
 243. Kawasaki C, Nishi R, Uekihara S, Hayano S, Otagiri M. Charcoal hemoperfusion in the treatment of phenytoin overdose. *Am J Kidney Dis*. 2000;35(2):323–326.
 244. Sung SF, Chiang PC, Tung HH, Ong CT. Charcoal hemoperfusion in an elderly man with life-threatening adverse reactions due to poor metabolism of phenytoin. *J Formos Med Assoc*. 2004;103(8):648–652.
 245. Ghannoum M, Troyanov S, Ayoub P, Laverigne V, Hewlett T. Successful hemodialysis in a phenytoin overdose: case report and review of the literature. *Clin Nephrol*. 2010;74(1):59–64.
 246. Frenchie D, Bastani B. Significant removal of phenytoin during high flux dialysis with cellulose triacetate dialyzer. *Nephrol Dial Transplant*. 1998;13(3):817–818.
 247. Cormier MJ, Desmeules S, St-Onge M, Ghannoum M. Phenytoin overdose treated with hemodialysis using a high cut-off dialyzer. *Hemodial Int*. 2017;21(1):E13–E17.
 248. Rubinger D, Levy M, Roll D, Czaczkes JW. Inefficiency of haemodialysis in acute phenytoin intoxication. *Br J Clin Pharmacol*. 1979;7(4):405–407.
 249. Czajka PA, Anderson WH, Christoph RA, Banner Jr W. A pharmacokinetic evaluation of peritoneal dialysis for phenytoin intoxication. *J Clin Pharmacol*. 1980;20(10):565–569.
 250. Lindahl S, Westerling D. Detoxification with peritoneal dialysis and blood exchange after diphenylhydantoin intoxication. *Acta Paediatr Scand*. 1982;71(4):665–666.
 251. Narcy P, Zorza G, Taburet AM, Mersch JM, Devictor D, Huault G. Severe poisoning with intravenous phenytoin in the newborn. Value of peritoneal dialysis. *Arch Fr Pediatr*. 1990;47(8):591–593.
 252. Lau AH, Kronfol NO. Effect of continuous hemofiltration on phenytoin elimination. *Ther Drug Monit*. 1994;16(1):53–57.
 253. Audia P, Feinfeld DA, Dubrow A, Winchester JF. Metformin-induced lactic acidosis and acute pancreatitis precipitated by diuretic, celecoxib, and candesartan-associated acute kidney dysfunction. *Clin Toxicol*. 2008;46(2):164–166.
 254. Chang CT, Chen YC, Fang JT, Huang CC. Metformin-associated lactic acidosis: case reports and literature review. *J Nephrol*. 2002;15(4):398–402.
 255. Peters N, Jay N, Barraud D, Cravoisy A, Nace L, Bollaert PE, et al. Metformin-associated lactic acidosis in an intensive care unit. *Crit Care*. 2008;12(6).
 256. Luft D, Deichsel G, Schmulling RM, Stein W, Eggstein M. Definition of clinically relevant lactic acidosis in patients with internal diseases. *Am J Clin Pathol*. 1983;80(4):484–489.
 257. Roberts DM, Ghannoum M. Expanding the evidence for managing metformin poisoning to support decision-making. *Clin Toxicol*. 2023;61(4):203–206.
 258. Seidowsky A, Nseir S, Houdret N, Fourrier F. Metformin-associated lactic acidosis: a prognostic and therapeutic study. *Crit Care Med*. 2009;37(7):2191–2196.
 259. Salpeter SR, Greyber E, Pasternak GA, Salpeter Posthumous EE. Risk of fatal and nonfatal lactic acidosis with metformin use in type 2 diabetes mellitus. *Cochrane Database Syst Rev*. 2010;(1):CD002967.
 260. Duong JK, Furlong TJ, Roberts DM, et al. The role of metformin in metformin-associated lactic acidosis (MALA): case series and formulation of a model of pathogenesis. *Drug Saf*. 2013.
 261. Wills BK, Bryant SM, Buckley P, Seo B. Can acute overdose of metformin lead to lactic acidosis? *Am J Emerg Med*. 2010;28(8):857–861.
 262. Aghabiklooei A, Mostafazadeh B, Shiva H. A fatal case of metformin intoxication. *Pakistan J Med Sci*. 2011;27(4):943–944.
 263. Arroyo AM, Walroth TA, Mowry JB, Kao LW. The MALAdy of metformin poisoning: is CVVH the cure? *Am J Ther*. 2010;17(1):96–100.
 264. Vecchio S, Giampreti A, Petrolini VM, et al. Metformin accumulation: lactic acidosis and high plasmatic metformin levels in a retrospective case series of 66 patients on chronic therapy. *Clin Toxicol*. 2014;52(2):129–135.
 265. Kajbaf F, Lalau JD. The prognostic value of blood pH and lactate and metformin concentrations in severe metformin-associated lactic acidosis. *BMC pharmacology & toxicology*. 2013;14:22.
 266. Juurlink DN, Roberts DM. The enigma of metformin-associated lactic acidosis. *Clin Toxicol*. 2014;52(2):85–87.
 267. Soyoral YU, Begenik H, Emre H, Aytemiz E, Ozturk M, Erkoç R. Dialysis therapy for lactic acidosis caused by metformin intoxication: presentation of two cases. *Hum Exp Toxicol*. 2011;30(12):1995–1997.
 268. van Berlo-van de Laar IRF, Vermeij CG, van den Elsen-Hutten M, de Meijer A, Taxis K, Jansman FGA. Extracorporeal treatment of metformin-associated lactic acidosis in clinical practice: a retrospective cohort study. *Eur J Clin Pharmacol*. 2020;76(6):815–820.
 269. Corchia A, Wynckel A, Journet J, et al. Metformin-related lactic acidosis with acute kidney injury: results of a French observational multicenter study. *Clin Toxicol*. 2020;58(5):375–382.
 270. Roberts D, Duong J, Ray J, Williams K, Furlong T. Therapeutic drug monitoring of metformin in a patient with end stage renal failure on haemodialysis. *Nephrology*. 2010;15:87–88.
 271. Barrueto F, Meggs WJ, Barchman MJ. Clearance of metformin by hemofiltration in overdose. *J Toxicol Clin Toxicol*. 2002;40(2):177–180.
 272. Lalau JD, Andrejak M, Moriniere P, et al. Hemodialysis in the treatment of lactic acidosis in diabetics treated by metformin: a study of metformin elimination. *Int J Clin Pharmacol Ther Toxicol*. 1989;27(6):285–288.
 273. Trebach J, Mohan S, Gnirke M, Su MK, Gosselin S, Hoffman RS. Retrospective evaluation of management guidelines for extracorporeal treatment of metformin poisoning. *Clin Toxicol*. 2023;61(4):223–227.
 274. Ayoub P, Héty PO, Cormier M, et al. Metformin toxicokinetics during hemodialysis. *Kidney int reports*. 2017;2:759–765.
 275. Storch S, Krimmerman S, Sabo E, et al. MID-high flux hemodialysis (HD) is equal to continuous venous venous hemodiafiltration (CVVHDF) in the treatment of severe lactic acidosis. *Intensive Care Med*. 2011;37:S130.
 276. Levraut J, Ciebiera JP, Jambou P, Ichai C, Labib Y, Grimaud D. Effect of continuous venovenous hemofiltration with dialysis on lactate clearance in critically ill patients. *Crit Care Med*. 1997;25(1):58–62.
 277. Stefani M, Roberts DM. Complex decisions in the use of extracorporeal treatments in acute metformin overdose: which modality, when and how to measure the effect. *Br J Clin Pharmacol*. 2018;84(12):2689–2691.
 278. Guo PY, Storsley LJ, Finkle SN. Severe lactic acidosis treated with prolonged hemodialysis: recovery after massive overdoses of metformin. *Semin Dial*. 2006;19(1):80–83.
 279. Kruse JA. Metformin-associated lactic acidosis. *J Emerg Med*. 2001;20(3):267–272.
 280. Gunnell D, Eddleston M, Phillips MR, Konradsen F. The global distribution of fatal pesticide self-poisoning: systematic review. *BMC Publ Health*. 2007;7:357.
 281. Smith LL. The toxicity of paraquat. *Adverse Drug Reactions and Acute Poisoning Review*. 1988;7(1):1–17.
 282. Houze P, Baud FJ, Mouy R, Bismuth C, Bourdon R, Scherrmann JM. Toxicokinetics of paraquat in humans. *Hum Exp Toxicol*. 1990;9(1):5–12.
 283. Bismuth C, Scherrmann JM, Garnier R. Elimination of paraquat. *Hum Toxicol*. 1987;6(1):63–67.

284. Bus JS, Gibson JE. Paraquat: model for oxidant-initiated toxicity. *Environ Health Perspect.* 1984;55:37–46.
285. Suntres ZE. Role of antioxidants in paraquat toxicity. *Toxicology.* 2002;180(1):65–77.
286. Bismuth C, Garnier R, Baud FJ, Muszynski J, Keyes C. Paraquat poisoning: an overview of the current status. *Drug Saf.* 1990;5(4):243–251.
287. Bismuth C, Garnier R, Dally S, Fournier PE, Scherrmann JM. Prognosis and treatment of paraquat poisoning: a review of 28 cases. *J Toxicol Clin Toxicol.* 1982;19(5):461–474.
288. Gil HW, Kang MS, Yang JO, Lee EY, Hong SY. Association between plasma paraquat level and outcome of paraquat poisoning in 375 paraquat poisoning patients. *Clin Toxicol.* 2008;46(6):515–518.
289. Vale JA, Meredith TJ, Buckley BM. Paraquat poisoning: clinical features and immediate general management. *Hum Toxicol.* 1987;6(1):41–47.
290. Rose MS, Smith LL, Wyatt I. Evidence for energy-dependent accumulation of paraquat into rat lung. *Nature.* 1974;252(5481):314–315.
291. Proudfoot AT, Stewart MS, Levitt T, Widdop B. Paraquat poisoning: significance of plasma-paraquat concentrations. *Lancet.* 1979;2(8138):330–332.
292. Scherrmann JM, Houze P, Bismuth C, Bourdon R. Prognostic value of plasma and urine paraquat concentration. *Hum Toxicol.* 1987;6(1):91–93.
293. Senarathna L, Eddleston M, Wilks MF, et al. Prediction of outcome after paraquat poisoning by measurement of the plasma paraquat concentration. *QJM.* 2009;102(4):251–259.
294. Ikebuchi J. Evaluation of paraquat concentrations in paraquat poisoning. *Arch Toxicol.* 1987;60(4):304–310.
295. Ikebuchi J, Proudfoot AT, Matsubara K, et al. Toxicological index of paraquat: a new strategy for assessment of severity of paraquat poisoning in 128 patients. *Forensic Sci Int.* 1993;59(2):85–87.
296. Lee EY, Hwang KY, Yang JO, Hong SY. Predictors of survival after acute paraquat poisoning. *Toxicol Ind Health.* 2002;18(4):201–206.
297. Gunawardena G, Roberts DM, Buckley NA. Randomized control trial of immunosuppression in paraquat poisoning. *Crit Care Med.* 2007;35(1):330–331; author reply 1.
298. Hampson EC, Effeney DJ, Pond SM. Efficacy of single or repeated hemoperfusion in a canine model of paraquat poisoning. *J Pharmacol Exp Ther.* 1990;254(2):732–740.
299. Koo JR, Yoon JW, Han SJ, et al. Rapid analysis of plasma paraquat using sodium dithionite as a predictor of outcome in acute paraquat poisoning. *Am J Med Sci.* 2009;338(5):373–377.
300. Bian X, Sung L, Jiang T. Hemoperfusion in the treatment of paraquat dichloride poisoning. *Hei Long Medical Journal.* 2006;30(10):796–797.
301. Alboni P, Menozzi C, Brignole M, et al. Effects of permanent pacemaker and oral theophylline in sick sinus syndrome the THEOPACE study: a randomized controlled trial. *Circulation.* 1997;96(1):260–266.
302. Hendeles L, Jenkins J, Temple R. Revised FDA labeling guideline for theophylline oral dosage forms. *Pharmacotherapy.* 1995;15(4):409–427.
303. Ogilvie RI. Clinical pharmacokinetics of theophylline. *Clin Pharmacokinet.* 1978;3(4):267–293.
304. Shannon M. Predictors of major toxicity after theophylline overdose. *Ann Intern Med.* 1993;119(12):1161–1167.
305. Sessler CN. Theophylline toxicity: clinical features of 116 consecutive cases. *Am J Med.* 1990;88(6):567–576.
306. Shannon M. Life-threatening events after theophylline overdose: a 10-year prospective analysis. *Arch Intern Med.* 1999;159(9):989–994.
307. Shannon M. Hypokalemia, hyperglycemia and plasma catecholamine activity after severe theophylline intoxication. *J Toxicol Clin Toxicol.* 1994;32(1):41–47.
308. Amin DN, Henry JA. Propranolol administration in theophylline overdose. *Lancet.* 1985;1(8427):520–521.
309. Seneff M, Scott J, Friedman B, Smith M. Acute theophylline toxicity and the use of esmolol to reverse cardiovascular instability. *Ann Emerg Med.* 1990;19(6):671–673.
310. Berlinger WG, Spector R, Goldberg MJ, Johnson GF, Quee CK, Berg MJ. Enhancement of theophylline clearance by oral activated charcoal. *Clin Pharmacol Ther.* 1983;33(3):351–354.
311. True RJ, Berman JM, Mahutte CK. Treatment of theophylline toxicity with oral activated charcoal. *Crit Care Med.* 1984;12(2):113–114.
312. Goldberg MJ, Spector R, Park GD, Johnson GF, Roberts P. The effect of sorbitol and activated charcoal on serum theophylline concentrations after slow-release theophylline. *Clin Pharmacol Ther.* 1987;41(1):108–111.
313. Woo OF, Pond SM, Benowitz NL, Olson KR. Benefit of hemoperfusion in acute theophylline intoxication. *J Toxicol Clin Toxicol.* 1984;22(5):411–424.
314. Biberstein MP, Ziegler MG, Ward DM. Use of beta-blockade and hemoperfusion for acute theophylline poisoning. *West J Med.* 1984;141(4):485–490.
315. Ehlers SM, Zaske DE, Sawchuk RJ. Massive theophylline overdose. Rapid elimination by charcoal hemoperfusion. *JAMA.* 1978;240(5):474–475.
316. Mydlik M, Derzsiova K, Smolko P, Petrikova V, Vzentekova T. Hemoperfusion with Amberlite XAD-4 in acute theophylline poisoning. *Cas Lek Cesk.* 1995;134(5):145–146.
317. Henderson JH, McKenzie CA, Hilton PJ, Leach RM. Continuous venovenous haemofiltration for the treatment of theophylline toxicity. *Thorax.* 2001;56(3):242–243.
318. Koeijers JJ, Verhoeven CL, Boersma HH, van Mook WNKA. Treatment of theophylline intoxication using continuous venovenous haemofiltration. *Neth J Crit Care.* 2008;12(1):32–34.
319. Osborn HH, Henry G, Wax P, Hoffman R, Howland MA. Theophylline toxicity in a premature neonate—elimination kinetics of exchange transfusion. *J Toxicol Clin Toxicol.* 1993;31(4):639–644.
320. Stocker ME, Montgomery JE. Serum paracetamol concentrations in adult volunteers following rectal administration. *Br J Anaesth.* 2001;87(4):638–640.
321. Hinson JA, Roberts DW, James LP. Mechanisms of acetaminophen-induced liver necrosis. *Handb Exp Pharmacol.* 2010;(196):369–405.
322. Larson AM, Polson J, Fontana RJ, et al. Acetaminophen-induced acute liver failure: results of a United States multicenter, prospective study. *Hepatology.* 2005;42(6):1364–1372.
323. Mazer M, Perrone J. Acetaminophen-induced nephrotoxicity: pathophysiology, clinical manifestations, and management. *J Med Toxicol.* 2008;4(1):2–6.
324. Rumack BH, Peterson RC, Koch GG, Amara IA. Acetaminophen overdose. 662 cases with evaluation of oral acetylcysteine treatment. *Arch Intern Med.* 1981;141(3 Spec No):380–385.
325. Hodgman MJ, Garrard AR. A review of acetaminophen poisoning. *Crit Care Clin.* 2012;28(4):499–516.
326. Green JL, Heard KJ, Reynolds KM, Albert D. Oral and intravenous acetylcysteine for treatment of acetaminophen toxicity: a systematic review and meta-analysis. *West J Emerg Med.* 2013;14(3):218–226.
327. Lin JH, Levy G. Sulfate depletion after acetaminophen administration and replenishment by infusion of sodium sulfate or N-acetylcysteine in rats. *Biochem Pharmacol.* 1981;30(19):2723–2725.
328. McGill MR, Sharpe MR, Williams CD, Taha M, Curry SC, Jaeschke H. The mechanism underlying acetaminophen-induced hepatotoxicity in humans and mice involves mitochondrial damage and nuclear DNA fragmentation. *J Clin Invest.* 2012;122(4):1574–1583.
329. Wu ML, Tsai WJ, Deng JF, Yang CC. Hemodialysis as adjunctive therapy for severe acetaminophen poisoning: a case report. *Zhonghua Yixue Zazhi.* 1999;62(12):907–913.
330. Chiew AL, Fountain JS, Graudins A, Isbister GK, Reith D, Buckley NA. Summary statement: new guidelines for the management of paracetamol poisoning in Australia and New Zealand. *Med J Aust.* 2015;203(5):215–218.
331. Zhou XM, Miao JY, Yang Y, et al. Clinical experience with molecular adsorbent recirculating system (MARS) in patients with drug-induced liver failure. *Artif Organs.* 2004;28(5):483–486.
332. de Geus H, Mathot R, van der Hoven B, Tjoa M, Bakker J. Enhanced paracetamol clearance with molecular adsorbents recirculating system (MARS(R)) in severe auto-intoxication. *Blood Purif.* 2010;30(2):118–119.
333. Karvellas CJ, Bagshaw SM, McDermid RC, Stollery DE, Gibney RT. Acetaminophen-induced acute liver failure treated with single-pass albumin dialysis: report of a case. *Int J Artif Organs.* 2008;31(5):450–455.
334. Detry O, Arkadopoulos N, Ting P, et al. Clinical use of a bioartificial liver in the treatment of acetaminophen-induced fulminant hepatic failure. *Am Surg.* 1999;65(10):934–938.

335. Ash SR, Caldwell CA, Singer GG, Lowell JA, Howard TK, Rustgi VK. Treatment of acetaminophen-induced hepatitis and fulminant hepatic failure with extracorporeal sorbent-based devices. *Adv Ren Replace Ther*. 2002;9(1):42–53.
336. Chan BS, Dawson AH, Buckley NA. What can clinicians learn from therapeutic studies about the treatment of acute oral methotrexate poisoning? *Clin Toxicol*. 2017;55(2):88–96.
337. Sand TE, Jacobsen S. Effect of urine pH and flow on renal clearance of methotrexate. *Eur J Clin Pharmacol*. 1981;19(6):453–456.
338. Bleyer WA. Methotrexate: clinical pharmacology, current status and therapeutic guidelines. *Cancer Treat Rev*. 1977;4(2):87–101.
339. Klemm A, Sperschneider H, Winnefeld K, Gunther K, Stein G. Comparison of aluminum removal between hemodialysis with polycarbonate low flux membrane and hemofiltration with polysulfone high flux membrane in end-stage renal failure patients. *Clin Nephrol*. 1997;47(2):133–134.
340. Bahlmann H, Lindwall R, Persson H. Acute barium nitrate intoxication treated by hemodialysis. *Acta Anaesthesiol Scand*. 2005;49(1):110–112.
341. Uchida HA, Sugiyama H, Kanehisa S, et al. An elderly patient with severe acute renal failure due to sodium bromate intoxication. *Intern Med*. 2006;45(3):151–154.
342. Auxiliadora-Martins M, Alkmin Teixeira GC, da Silva GS, et al. Severe encephalopathy after ingestion of star fruit juice in a patient with chronic renal failure admitted to the intensive care unit. *Heart Lung*. 2010;39(5):448–452.
343. Bresson J, Paugam-Burtz C, Josserand J, Bardin C, Mantz J, Pease S. Cefepime overdosage with neurotoxicity recovered by high-volume haemofiltration. *J Antimicrob Chemother*. 2008;62(4):849–850.
344. Singh T, Maw TT, Henry BL, et al. Extracorporeal therapy for dabigatran removal in the treatment of acute bleeding: a single center experience. *Clin J Am Soc Nephrol*. 2013;8(9):1533–1539.
345. Ghannoum M, Nolin TD. What is the role of renal replacement therapy in the setting of dabigatran toxicity? *Semin Dial*. 2014;27(3):223–226.
346. Ghannoum M, Cormier M, Bernier-Jean A, Brindamour D, Deziel C, Bouchard J. Extracorporeal treatments in a dapsone overdose: a case report. *Clin Toxicol*. 2016;54(9):886–889.
347. Brophy PD, Tenenbein M, Gardner J, Bunchman TE, Smoyer WE. Childhood diethylene glycol poisoning treated with alcohol dehydrogenase inhibitor fomepizole and hemodialysis. *Am J Kidney Dis*. 2000;35(5):958–962.
348. Usuda K, Kono K, Watanabe T, et al. Hemodialyzability of ionizable fluoride in hemodialysis session. *Sci Total Environ*. 2002;297(1–3):183–191.
349. Lu CM, James SH, Lien YH. Acute massive gentamicin intoxication in a patient with end-stage renal disease. *Am J Kidney Dis*. 1996;28(5):767–771.
350. Zosel A, Egelhoff E, Heard K. Severe lactic acidosis after an iatrogenic propylene glycol overdose. *Pharmacotherapy*. 2010;30(2):219.
351. Klansuwan N, Ratanajamit C, Kasiwong S, Wangsiripaisan A. Clearance of vancomycin during high-efficiency hemodialysis. *J Med Assoc Thai*. 2006;89(7):986–991.

QUESTIONS

1. Which of the following extracorporeal treatment provides the highest clearance for toxic alcohols?
 - a. Intermittent hemodialysis
 - b. Exchange transfusion
 - c. Continuous renal replacement therapy (CRRT)
 - d. Plasmapheresis
 - e. Extracorporeal liver assist devices

Answer: a

Rationale: Hemodialysis provides the highest clearance for all poisons aside from those that are extensively protein bound, partly because of high achievable blood and effluent flow.

2. For which of the following poisons does extracorporeal treatments ECTR have the lowest effect on total clearance?
 - a. Carbamazepine
 - b. Lithium
 - c. Amitriptyline
 - d. Phenobarbital

Answer: c

Rationale: Extracorporeal removal is not effective due to the large volume of distribution and high clearance of amitriptyline.

3. Why has the use of hemoperfusion in poisonings decreased over the years?
 - a. It is more expensive than hemodialysis.
 - b. Cartridges are not as available.
 - c. It has a greater complication rate than hemodialysis.
 - d. The advent of new dialyzers has rendered dialysis at least as efficient as hemoperfusion for most poisonings.
 - e. All the above.

Answer: e

Rationale: Hemoperfusion cartridges also need to be changed every 2 to 3 hours because of saturation. Finally, hemoperfusion cannot correct electrolyte disorders and does not bind metals or alcohols.